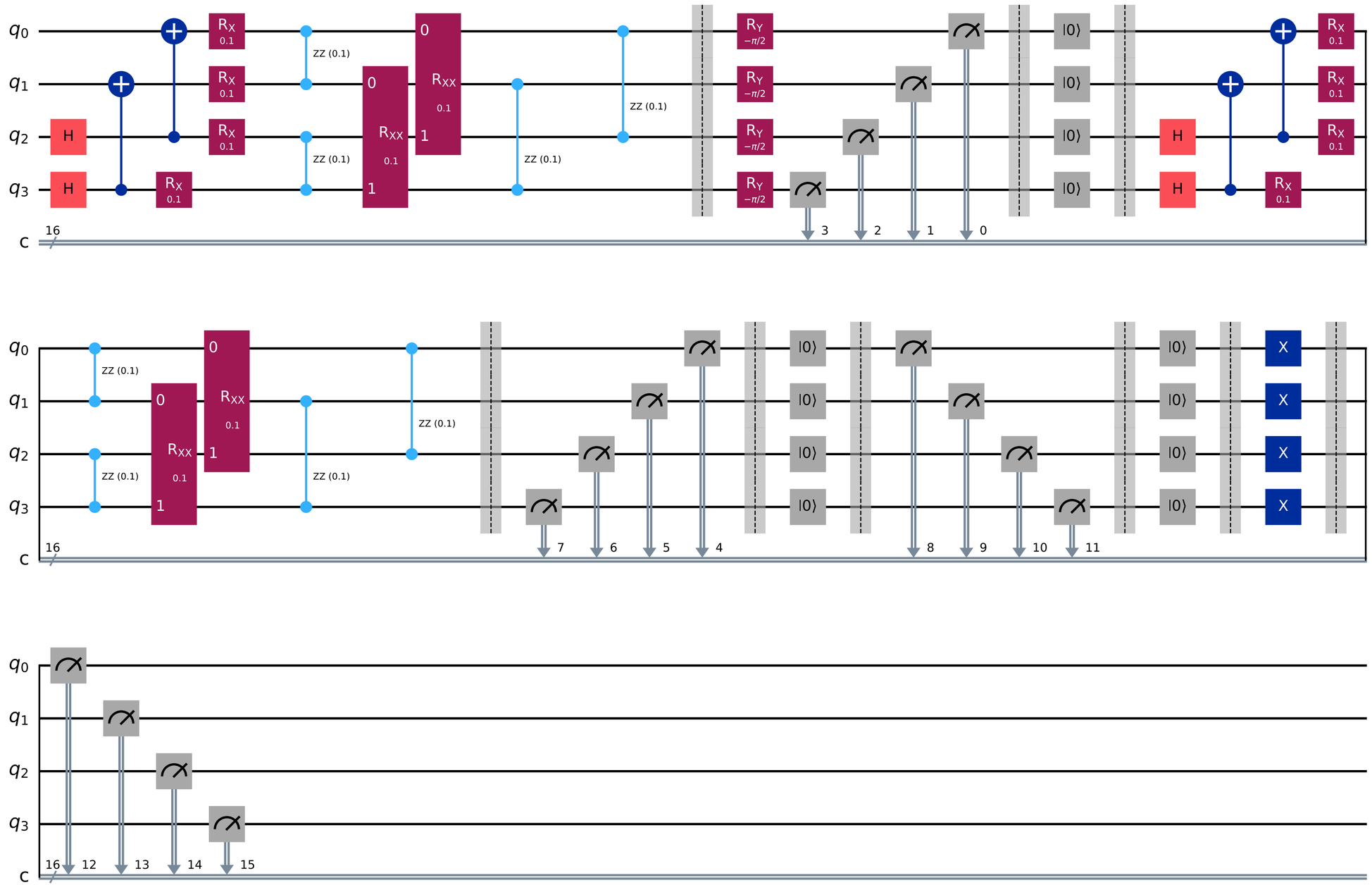
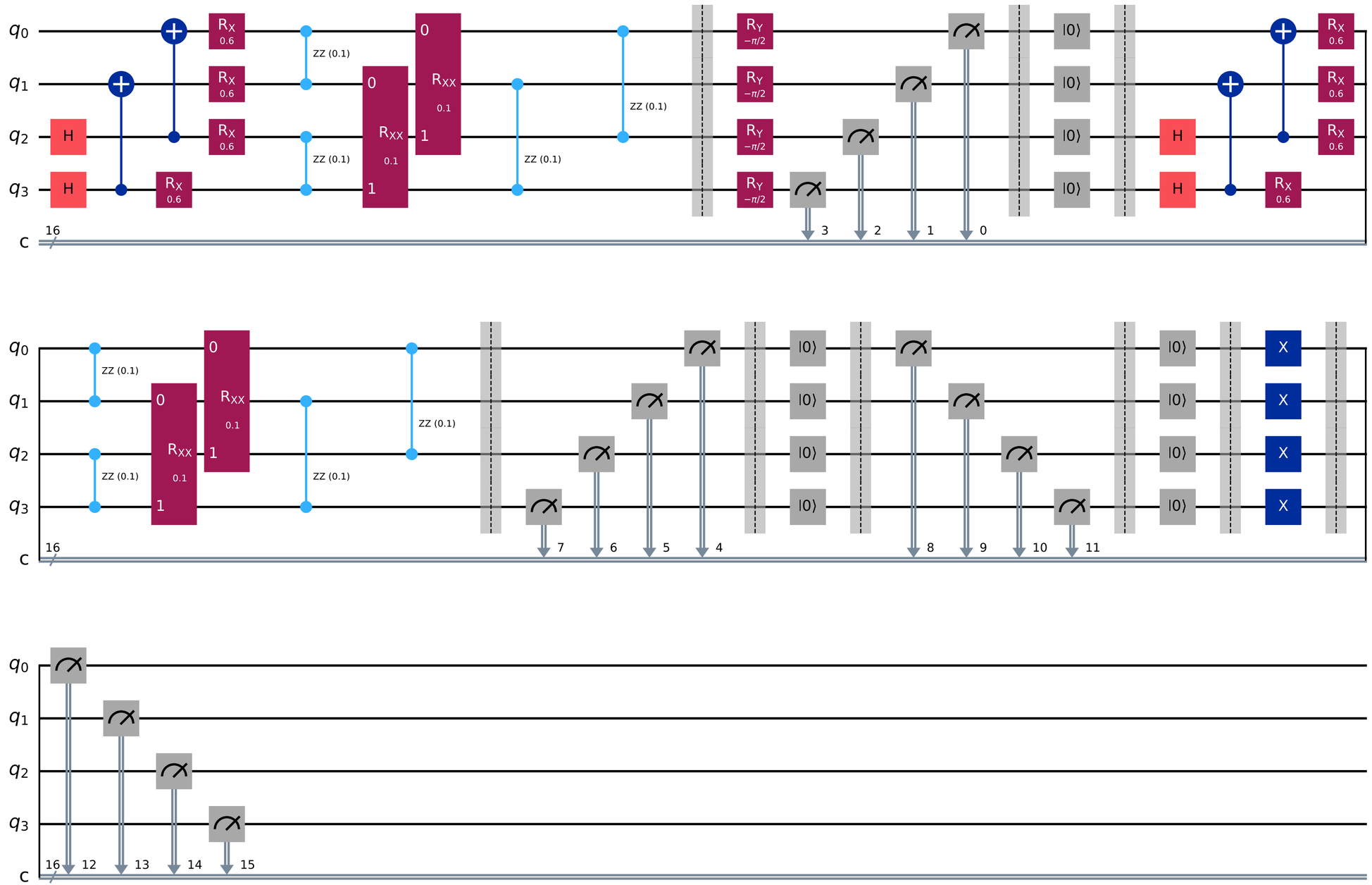
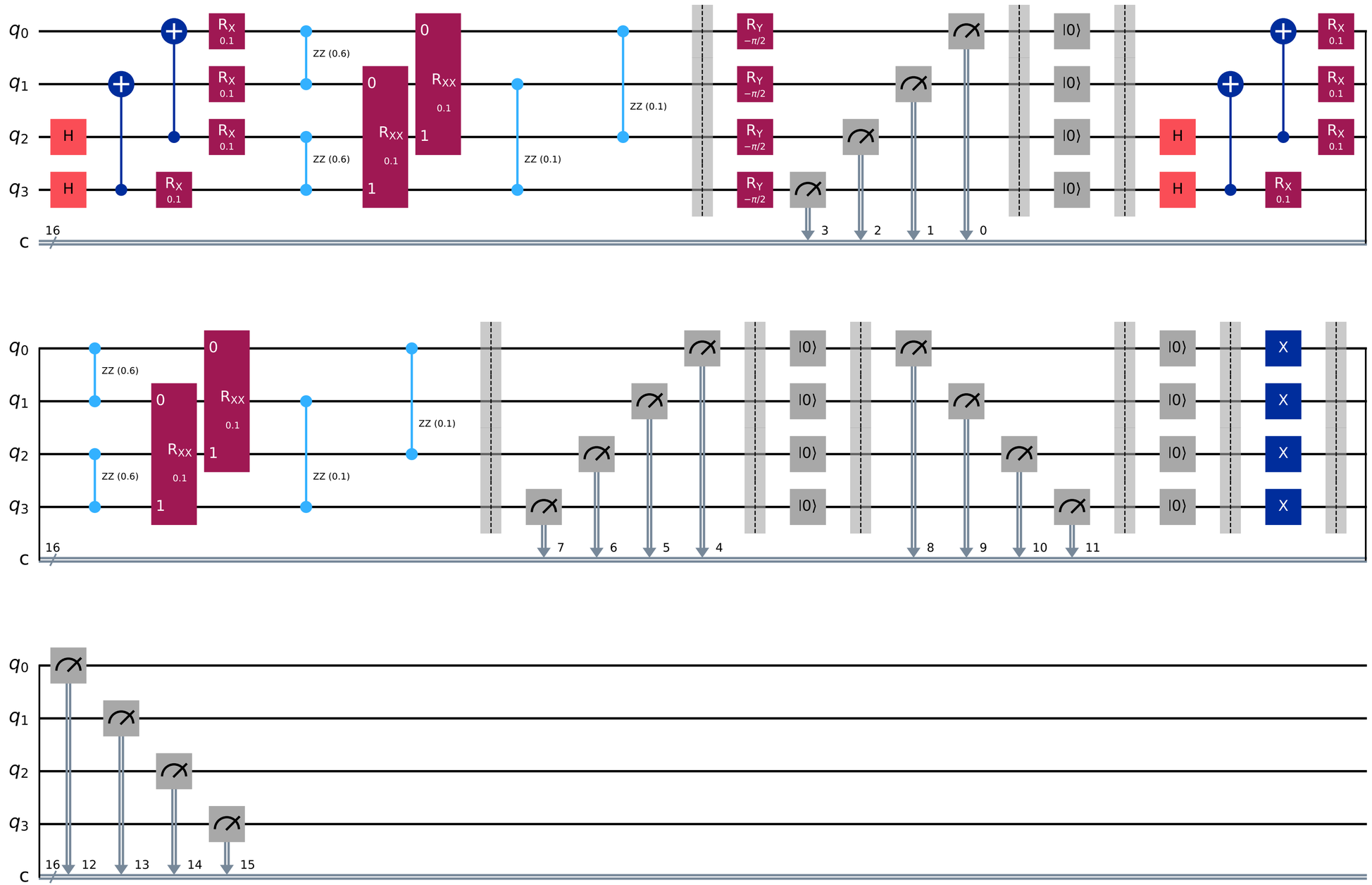
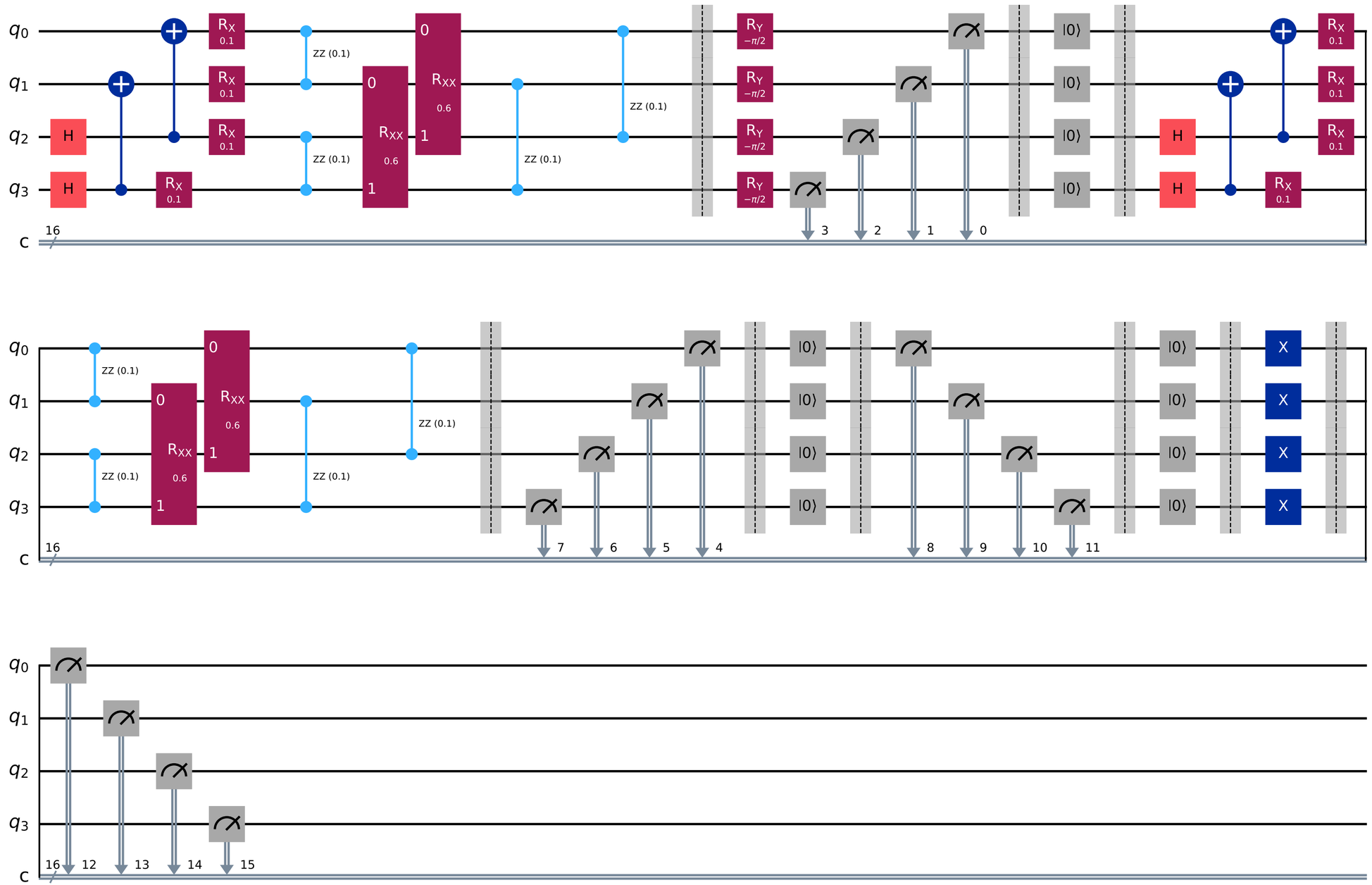
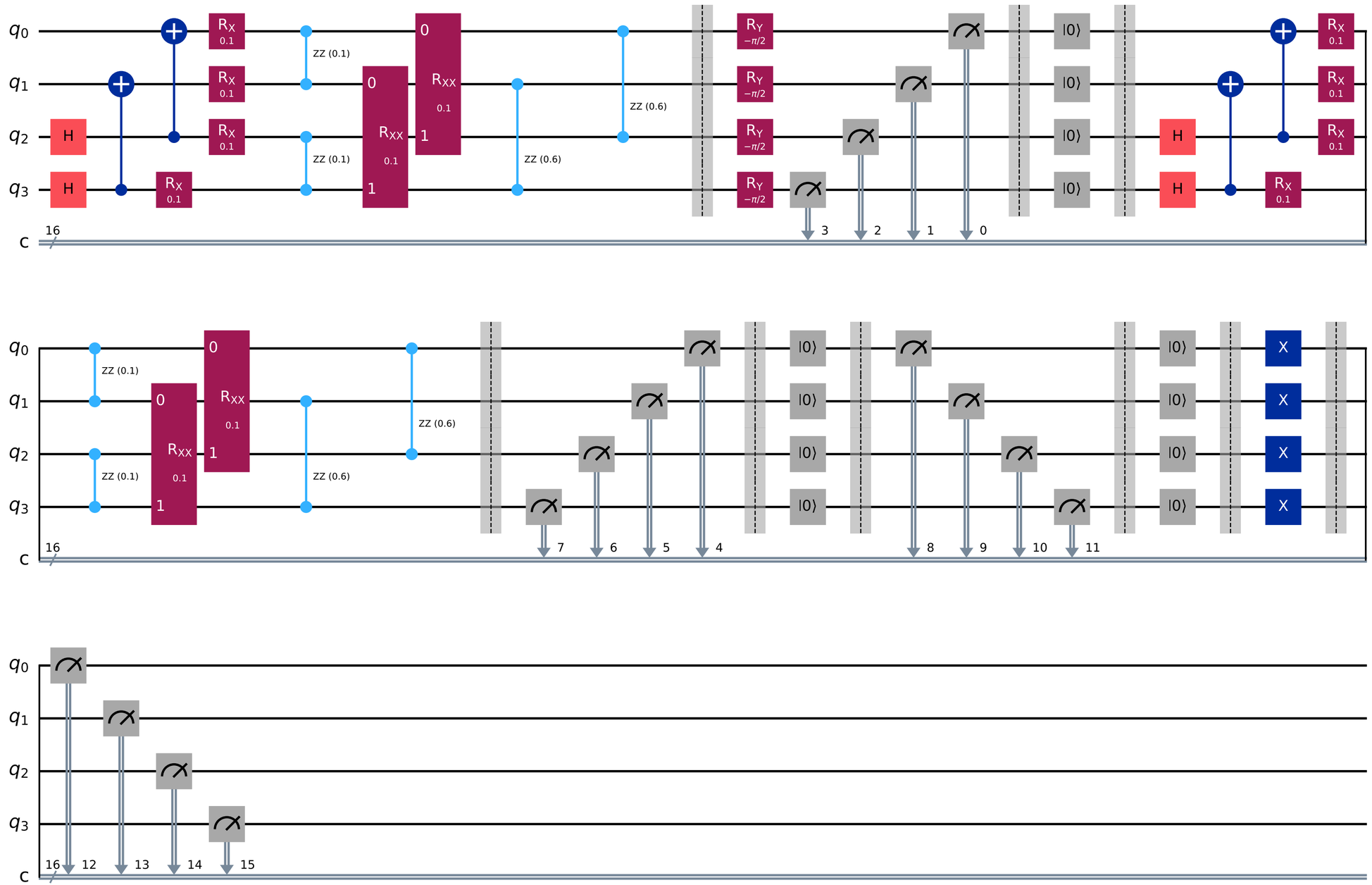
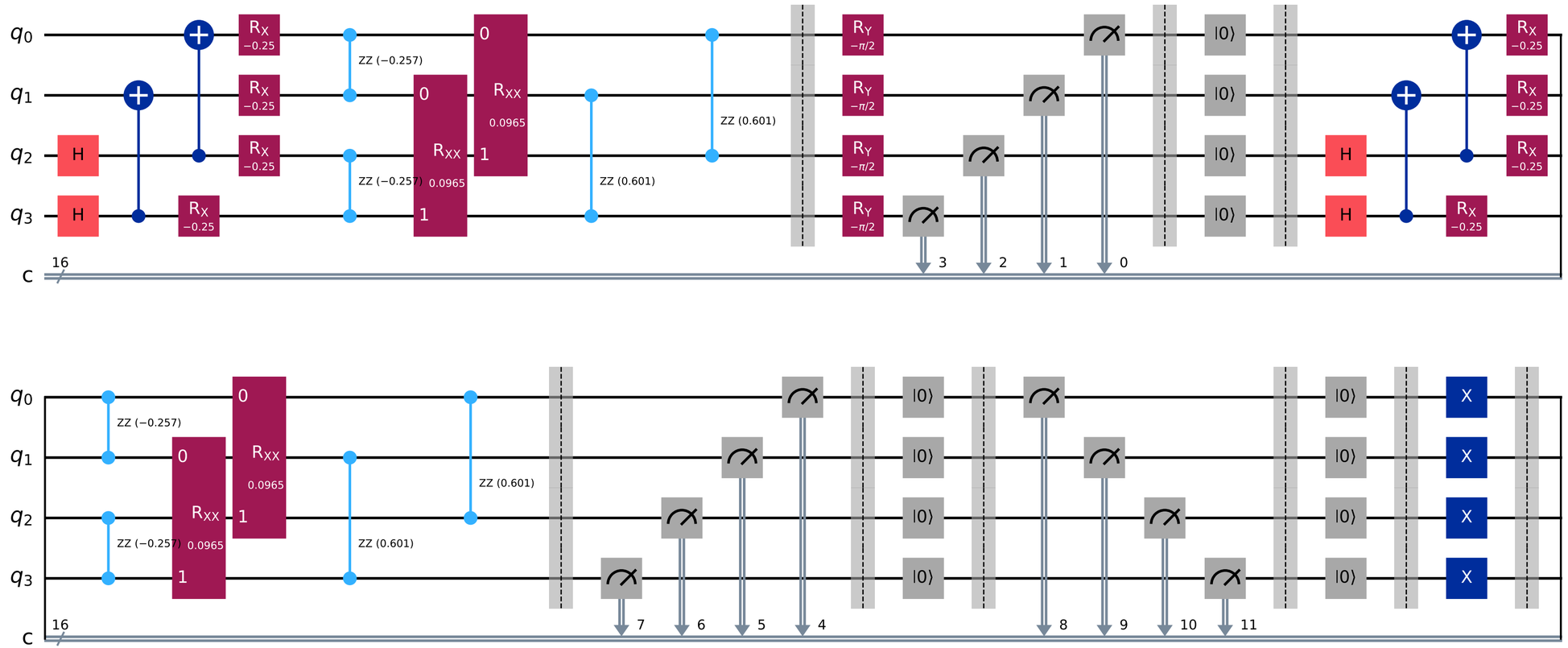

q_0
q_1
q_2
q_3
c
16
H
H
R_X 0.1
R_{XX} 0.1
ZZ (0.1)
R_Y $-\pi/2$
|0⟩
X
3
2
1
0
4
5
6
7
8
9
10
11
12
13
14
15

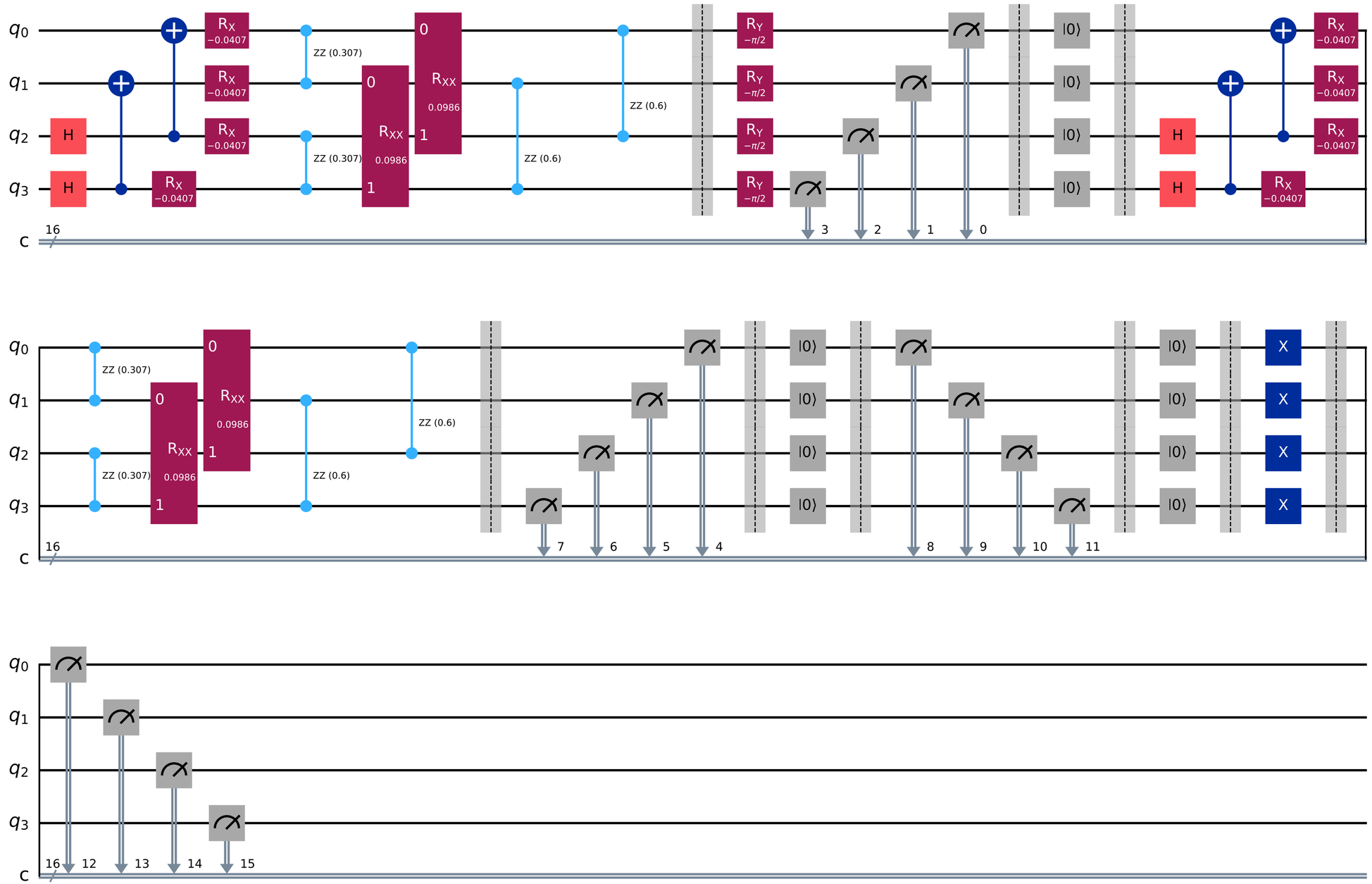

q_0
q_1
q_2
q_3
c
16
H
H
R_X 0.6
R_{XX} 0.1
ZZ (0.1)
R_Y $-\pi/2$
|0⟩
X
3
2
1
0
7
6
5
4
8
9
10
11
12
13
14
15

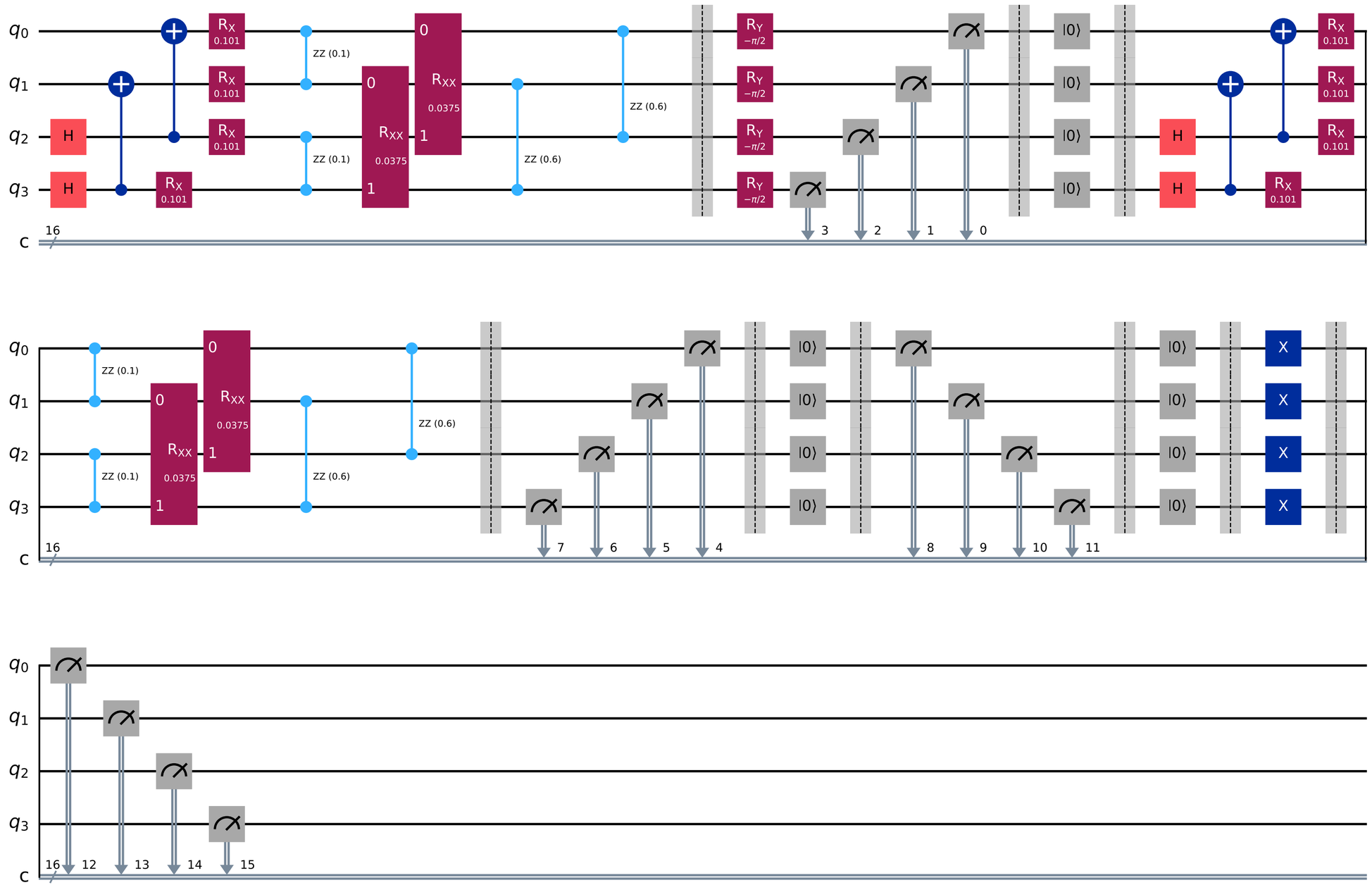

q_0
q_1
q_2
q_3
c
16
H
H
R_X 0.1
R_{XX} 0.1
ZZ (0.6)
ZZ (0.1)
R_Y $-\pi/2$
|0⟩
X
3
2
1
0
4
5
6
7
8
9
10
11
12
13
14
15

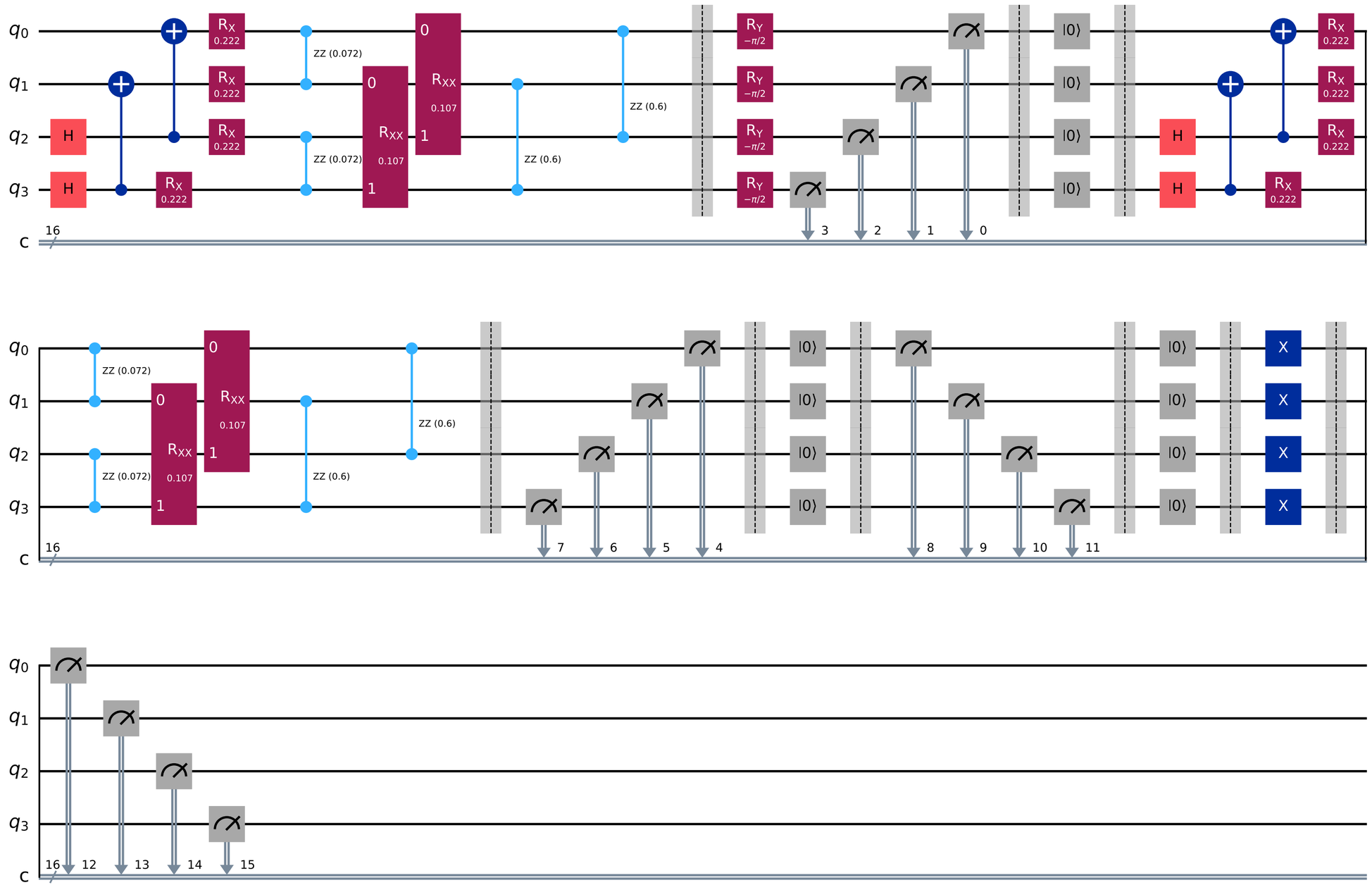

q_0
q_1
q_2
q_3
c
16
H
H
R_X 0.1
R_X 0.1
R_X 0.1
R_X 0.1
ZZ (0.1)
ZZ (0.1)
0
R_{XX} 0.6
1
0
R_{XX} 0.6
1
ZZ (0.1)
ZZ (0.1)
R_Y $-\pi/2$
R_Y $-\pi/2$
R_Y $-\pi/2$
R_Y $-\pi/2$
3
2
1
0
|0⟩
|0⟩
|0⟩
|0⟩
H
H
R_X 0.1
R_X 0.1
R_X 0.1
R_X 0.1
ZZ (0.1)
ZZ (0.1)
0
R_{XX} 0.6
1
0
R_{XX} 0.6
1
ZZ (0.1)
ZZ (0.1)
7
6
5
4
|0⟩
|0⟩
|0⟩
|0⟩
8
9
10
11
|0⟩
|0⟩
|0⟩
|0⟩
X
X
X
X
12
13
14
15

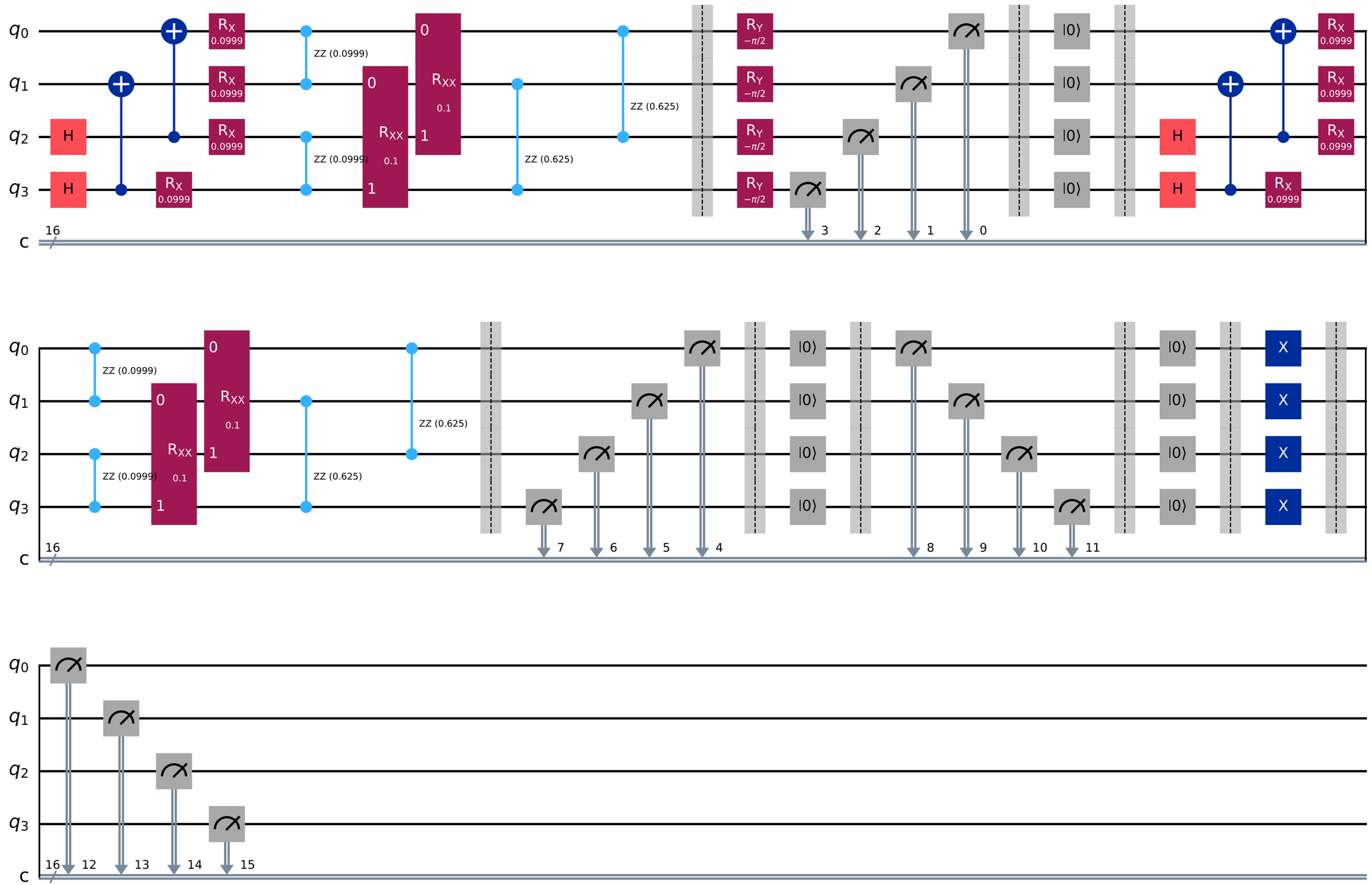

q_0
q_1
q_2
q_3
c
16
H
H
R_X 0.1
R_{XX} 0.1
ZZ (0.1)
ZZ (0.6)
R_Y $-\pi/2$
|0⟩
X
3
2
1
0
7
6
5
4
8
9
10
11
12
13
14
15

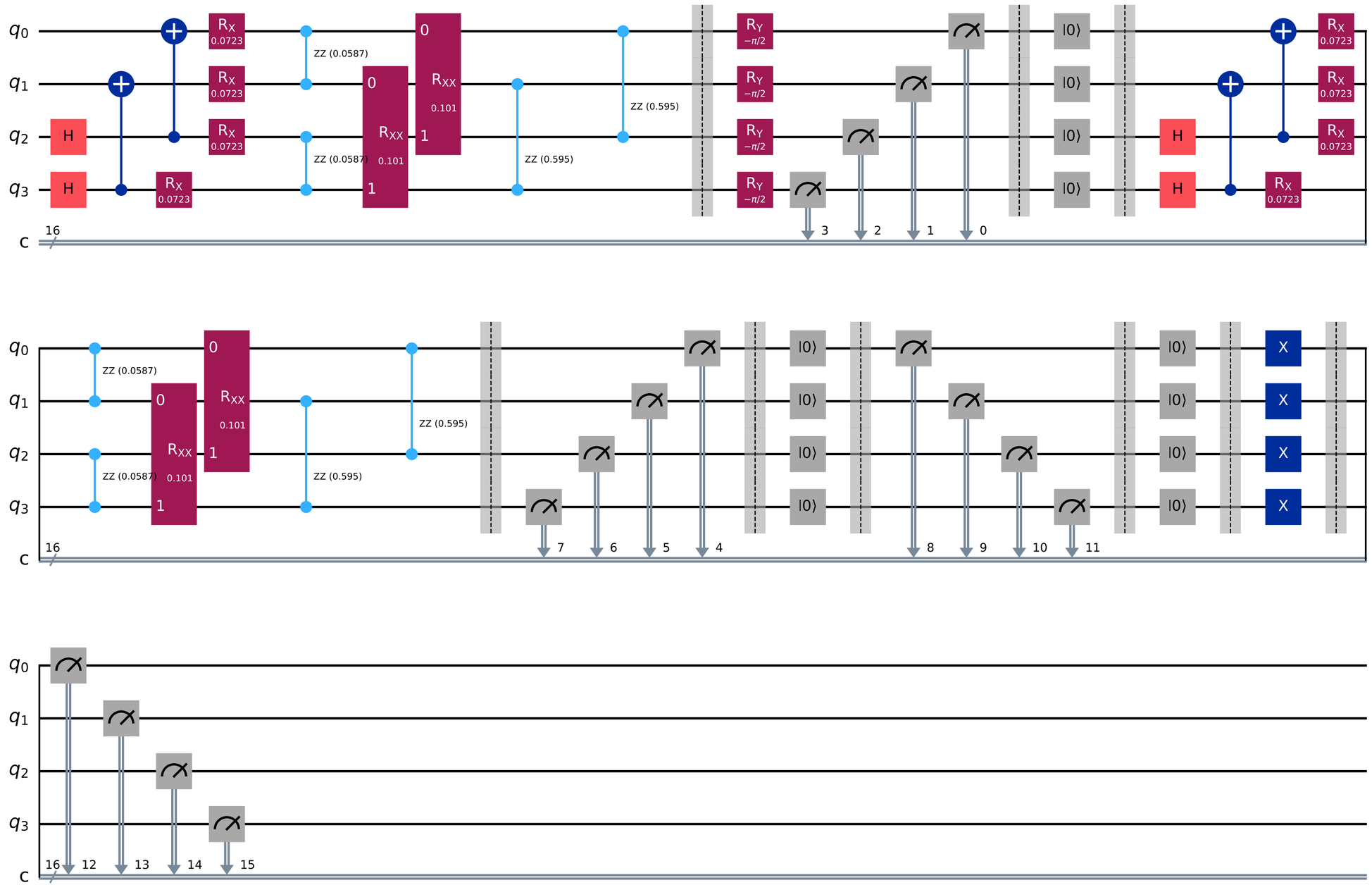

q_0
q_1
q_2
q_3
c
16
H
H
R_X −0.25
R_X −0.25
R_X −0.25
R_X −0.25
ZZ (−0.257)
ZZ (−0.257)
0
R_{XX}
0.0965
1
ZZ (0.601)
ZZ (0.601)
R_Y −π/2
R_Y −π/2
R_Y −π/2
R_Y −π/2
3
2
1
0
|0⟩
|0⟩
|0⟩
|0⟩
H
H
R_X −0.25
R_X −0.25
R_X −0.25
R_X −0.25
ZZ (−0.257)
ZZ (−0.257)
0
R_{XX}
0.0965
1
ZZ (0.601)
ZZ (0.601)
7
6
5
4
|0⟩
|0⟩
|0⟩
|0⟩
8
9
10
11
|0⟩
|0⟩
|0⟩
|0⟩
X
X
X
X
12
13
14
15

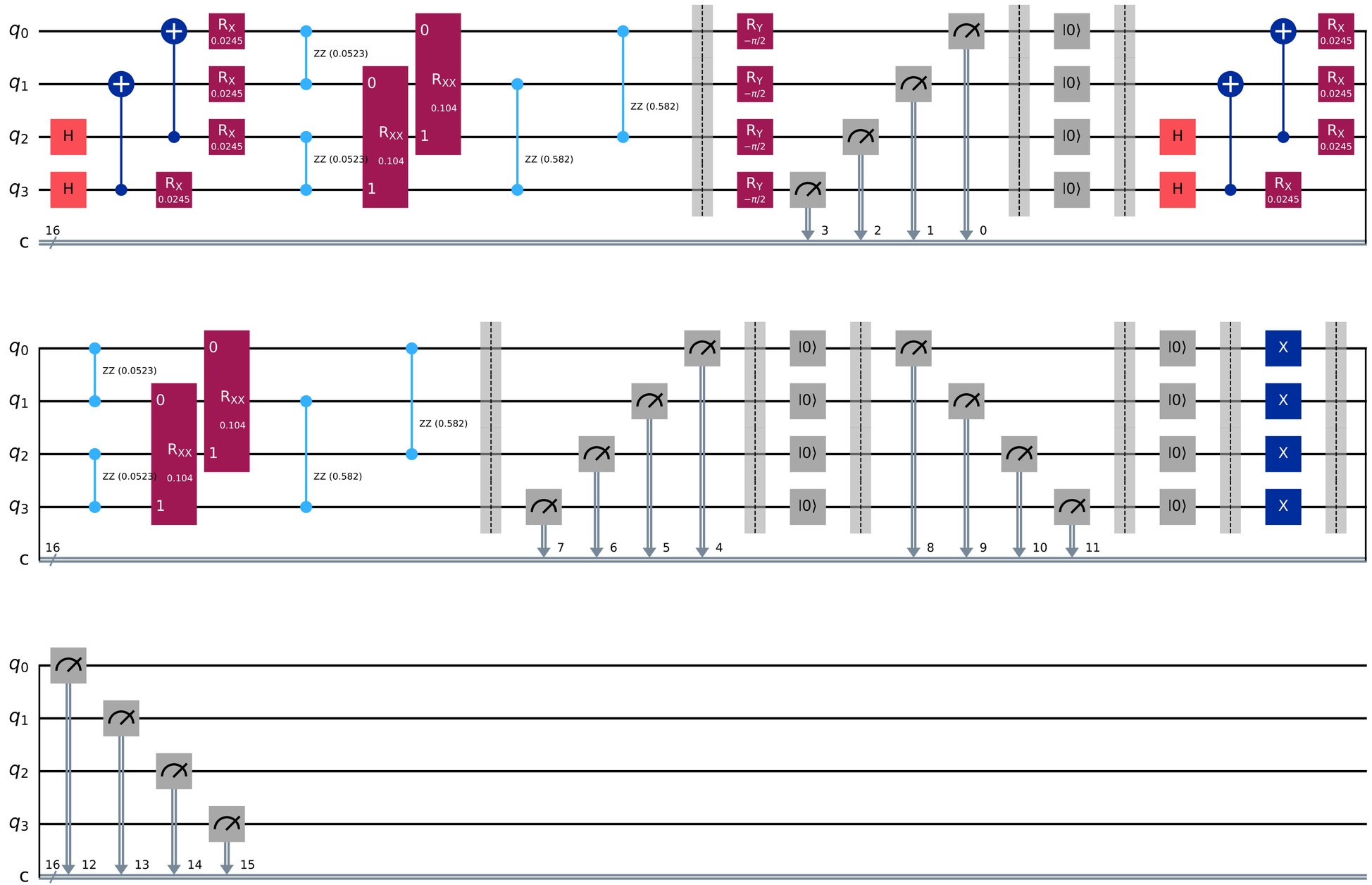

q0
q1
q2
q3
c
16
H
H
RX −0.0407
RX −0.0407
RX −0.0407
RX −0.0407
ZZ (0.307)
ZZ (0.307)
0
RXX
0.0986
1
0
RXX
0.0986
1
ZZ (0.6)
ZZ (0.6)
RY −π/2
RY −π/2
RY −π/2
RY −π/2
3
2
1
0
|0⟩
|0⟩
|0⟩
|0⟩
H
H
RX −0.0407
RX −0.0407
RX −0.0407
RX −0.0407
ZZ (0.307)
ZZ (0.307)
0
RXX
0.0986
1
ZZ (0.6)
ZZ (0.6)
7
6
5
4
8
9
10
11
|0⟩
|0⟩
|0⟩
|0⟩
|0⟩
X
X
X
X
12
13
14
15

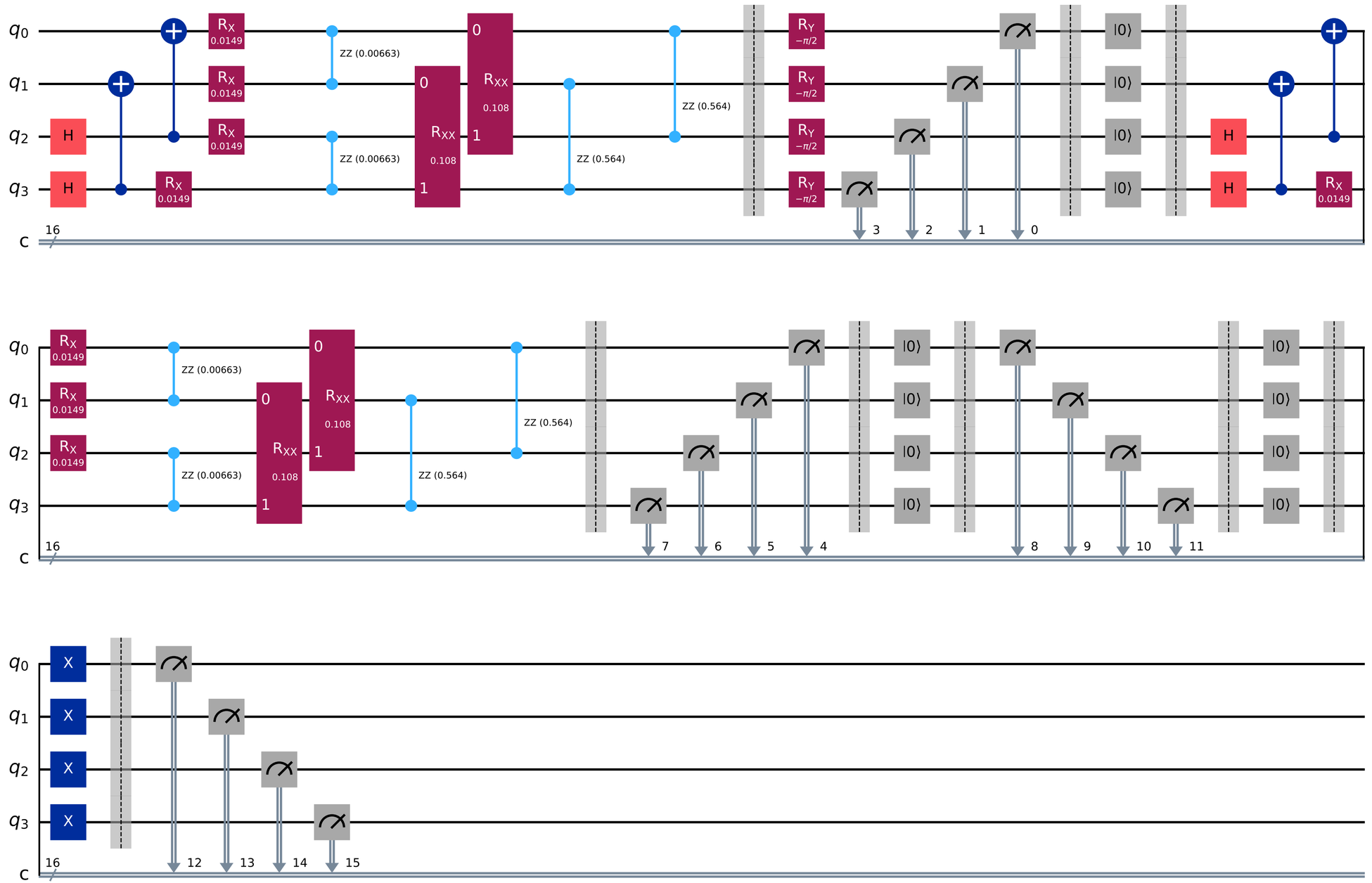

q_0
q_1
q_2
q_3
c
16
H
H
R_X 0.101
R_X 0.101
R_X 0.101
R_X 0.101
ZZ (0.1)
ZZ (0.1)
R_{XX} 0.0375
R_{XX} 0.0375
ZZ (0.6)
ZZ (0.6)
R_Y $-\pi/2$
R_Y $-\pi/2$
R_Y $-\pi/2$
R_Y $-\pi/2$
3
2
1
0
|0⟩
|0⟩
|0⟩
|0⟩
H
H
R_X 0.101
R_X 0.101
R_X 0.101
R_X 0.101
ZZ (0.1)
ZZ (0.1)
R_{XX} 0.0375
R_{XX} 0.0375
ZZ (0.6)
ZZ (0.6)
7
6
5
4
|0⟩
|0⟩
|0⟩
|0⟩
8
9
10
11
|0⟩
|0⟩
|0⟩
|0⟩
X
X
X
X
12
13
14
15

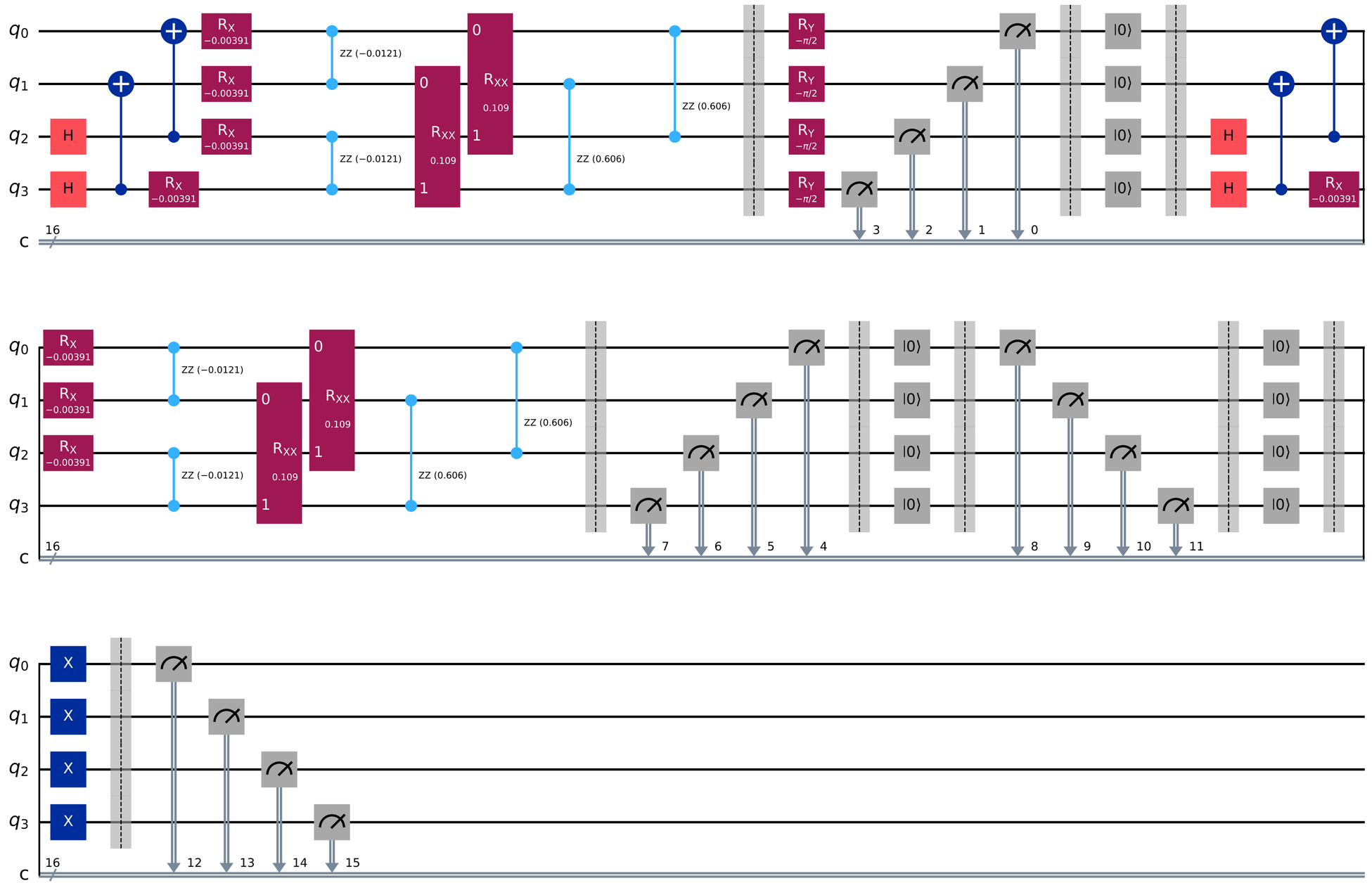

q_0
q_1
q_2
q_3
c
16
H
H
R_X 0.222
R_X 0.222
R_X 0.222
R_X 0.222
ZZ (0.072)
ZZ (0.072)
0
R_{XX} 0.107
1
ZZ (0.6)
ZZ (0.6)
R_Y $-\pi/2$
R_Y $-\pi/2$
R_Y $-\pi/2$
R_Y $-\pi/2$
3
2
1
0
|0⟩
|0⟩
|0⟩
|0⟩
H
H
R_X 0.222
R_X 0.222
R_X 0.222
R_X 0.222
ZZ (0.072)
ZZ (0.072)
0
R_{XX} 0.107
1
ZZ (0.6)
ZZ (0.6)
7
6
5
4
|0⟩
|0⟩
|0⟩
|0⟩
8
9
10
11
|0⟩
|0⟩
X
X
X
X
12
13
14
15

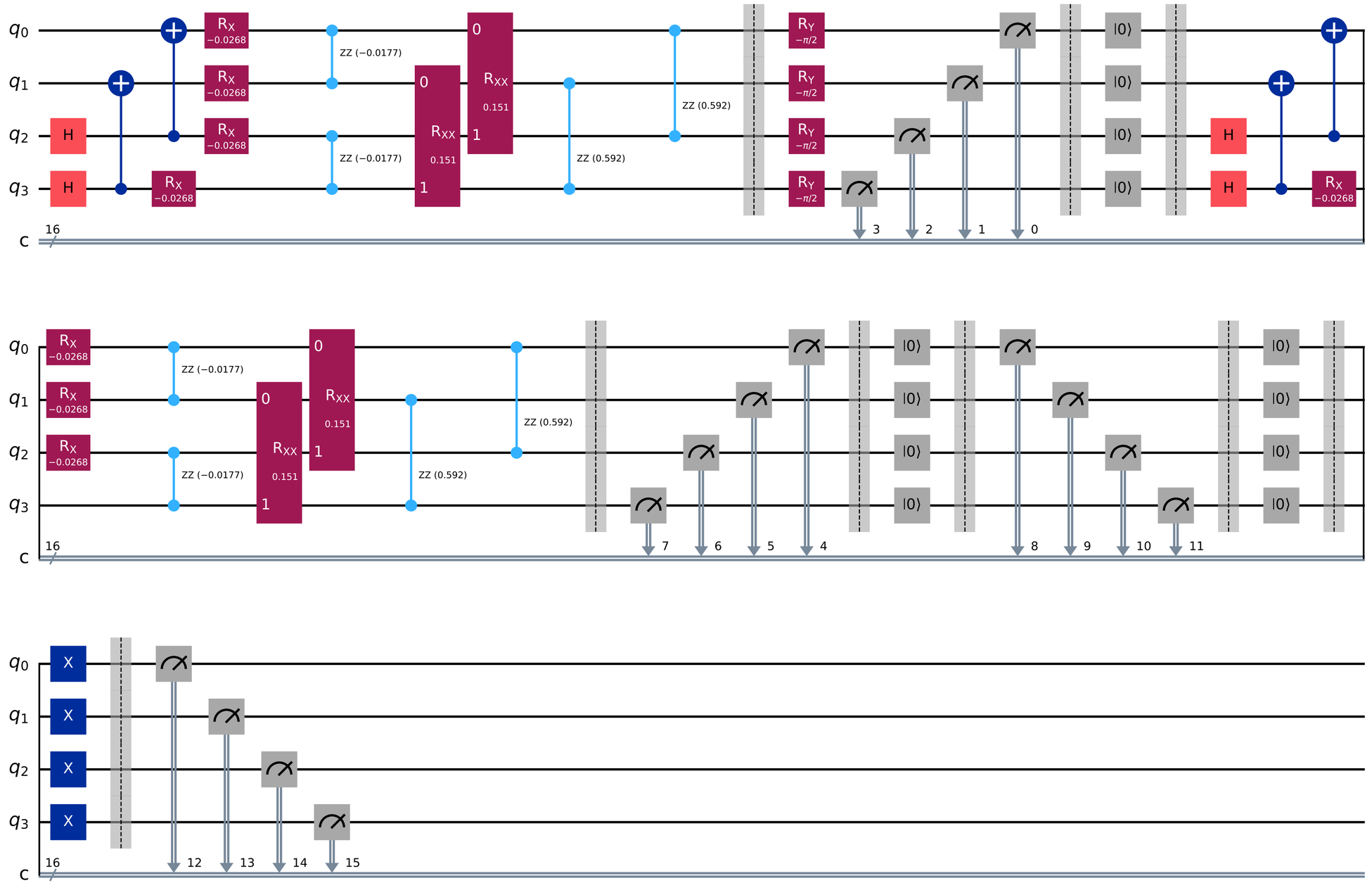

q0
q1
q2
q3
c
16
H
H
RX 0.0999
RX 0.0999
RX 0.0999
RX 0.0999
ZZ (0.0999)
ZZ (0.0999)
RXX 0.1
ZZ (0.625)
ZZ (0.625)
RY −π/2
RY −π/2
RY −π/2
RY −π/2
3
2
1
0
|0⟩
|0⟩
|0⟩
|0⟩
H
H
RX 0.0999
RX 0.0999
RX 0.0999
RX 0.0999
ZZ (0.0999)
ZZ (0.0999)
RXX 0.1
ZZ (0.625)
ZZ (0.625)
7
6
5
4
8
9
10
11
|0⟩
|0⟩
|0⟩
|0⟩
|0⟩
|0⟩
X
X
X
X
12
13
14
15

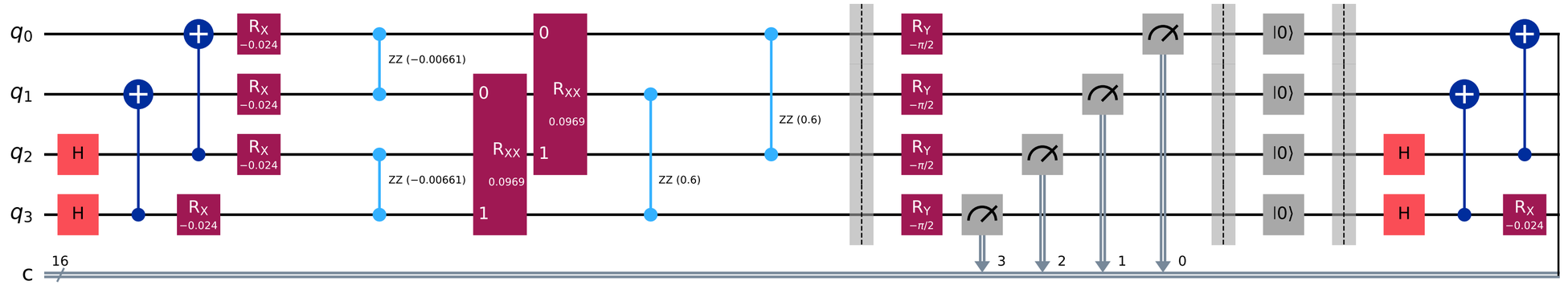

q0
q1
q2
q3
c
16
H
H
RX 0.0723
RX 0.0723
RX 0.0723
RX 0.0723
ZZ (0.0587)
ZZ (0.0587)
0
RXX 0.101
1
ZZ (0.595)
ZZ (0.595)
RY −π/2
RY −π/2
RY −π/2
RY −π/2
3
2
1
0
|0⟩
|0⟩
|0⟩
|0⟩
H
H
RX 0.0723
RX 0.0723
RX 0.0723
RX 0.0723
ZZ (0.0587)
ZZ (0.0587)
0
RXX 0.101
1
ZZ (0.595)
ZZ (0.595)
7
6
5
4
|0⟩
|0⟩
|0⟩
|0⟩
8
9
10
11
|0⟩
|0⟩
X
X
X
X
12
13
14
15

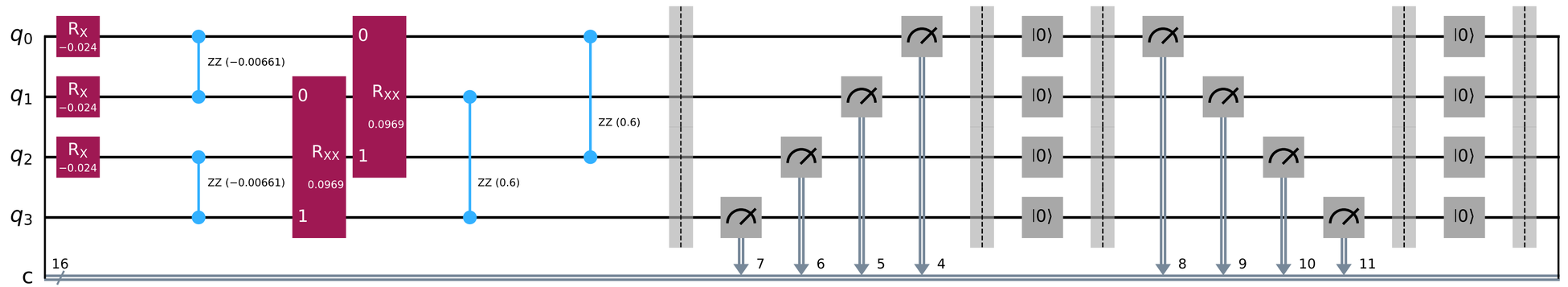

q0
q1
q2
q3
c
16
H
H
RX 0.0245
RX 0.0245
RX 0.0245
RX 0.0245
ZZ (0.0523)
ZZ (0.0523)
0
RXX 0.104
1
ZZ (0.582)
ZZ (0.582)
RY −π/2
RY −π/2
RY −π/2
RY −π/2
3
2
1
0
|0⟩
|0⟩
|0⟩
|0⟩
H
H
RX 0.0245
RX 0.0245
RX 0.0245
RX 0.0245
ZZ (0.0523)
ZZ (0.0523)
0
RXX 0.104
1
ZZ (0.582)
ZZ (0.582)
7
6
5
4
|0⟩
|0⟩
|0⟩
|0⟩
8
9
10
11
|0⟩
|0⟩
X
X
X
X
16
12
13
14
15

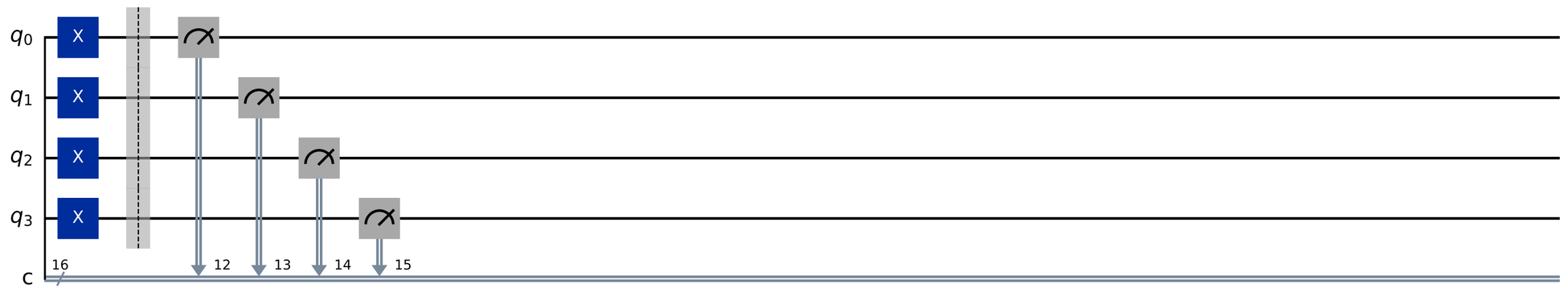

q0
q1
q2
q3
c
16
H
H
RX 0.0149
ZZ (0.00663)
ZZ (0.00663)
RXX 0.108
ZZ (0.564)
ZZ (0.564)
RY −π/2
|0⟩
3
2
1
0
X
12
13
14
15

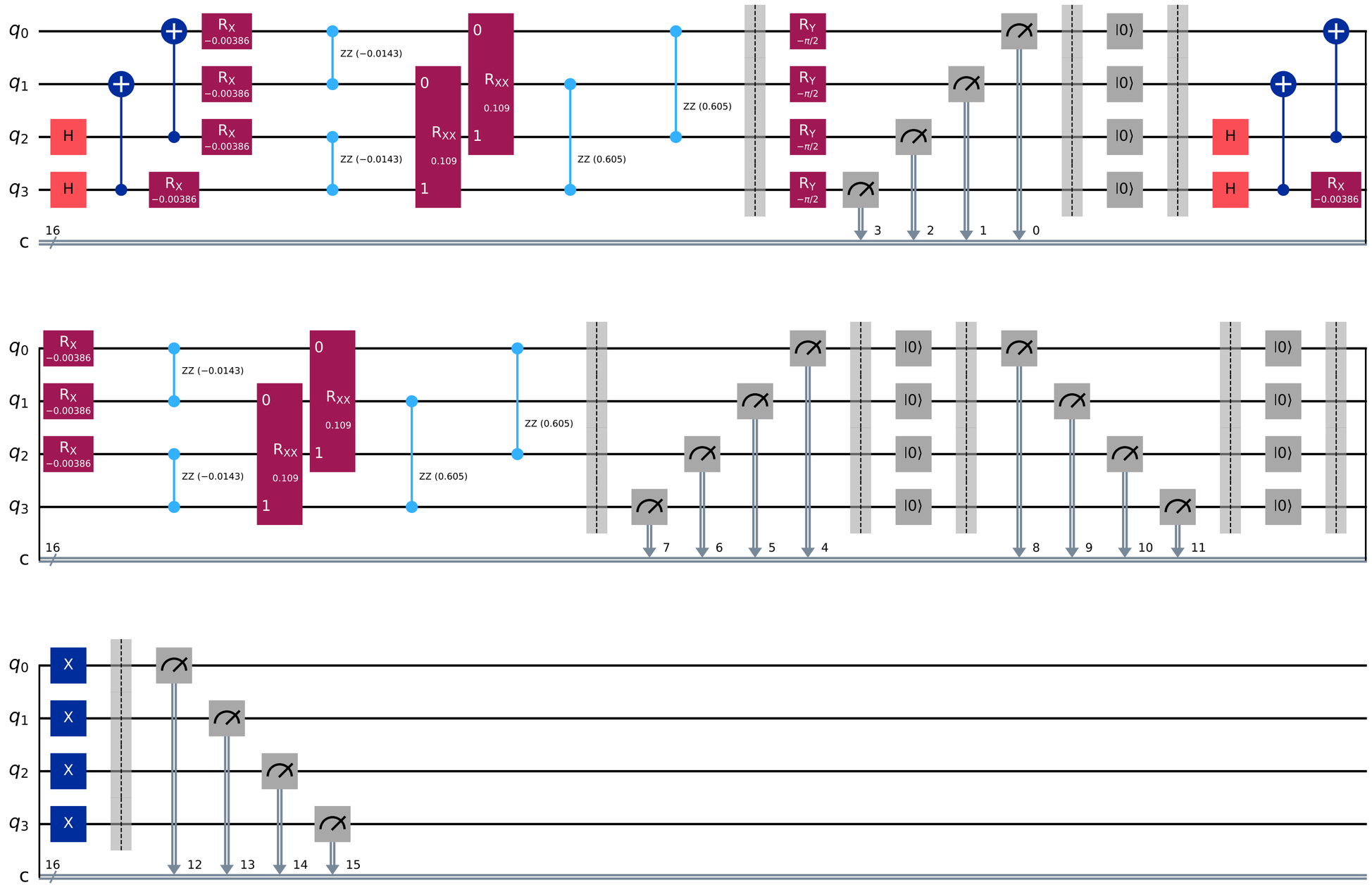

q_0
q_1
q_2
q_3
c
16
H
H
R_X −0.00391
ZZ (−0.0121)
ZZ (−0.0121)
0
R_{XX}
0.109
1
ZZ (0.606)
ZZ (0.606)
R_Y −π/2
|0⟩
3
2
1
0
7
6
5
4
8
9
10
11
X
12
13
14
15


q_0
q_1
q_2
q_3
c
16
H
H
R_X −0.0268
ZZ (−0.0177)
ZZ (0.592)
R_{XX} 0.151
R_Y −π/2
|0⟩
X
3
2
1
0
7
6
5
4
8
9
10
11
12
13
14
15

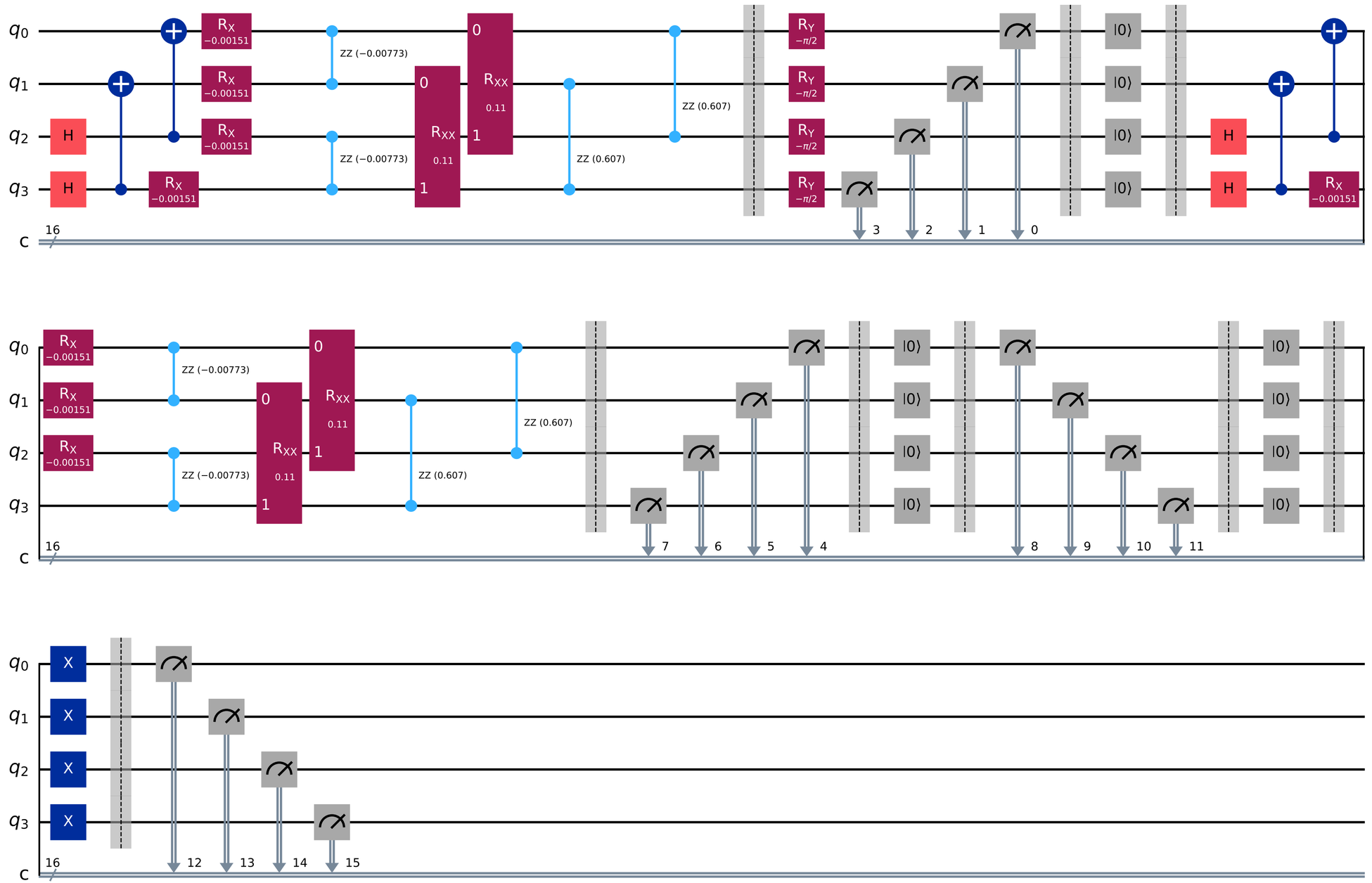

q_0
q_1
q_2
q_3
c
16
H
H
R_X −0.024
ZZ (−0.00661)
ZZ (0.6)
R_{XX} 0.0969
R_Y −π/2
|0⟩
3
2
1
0

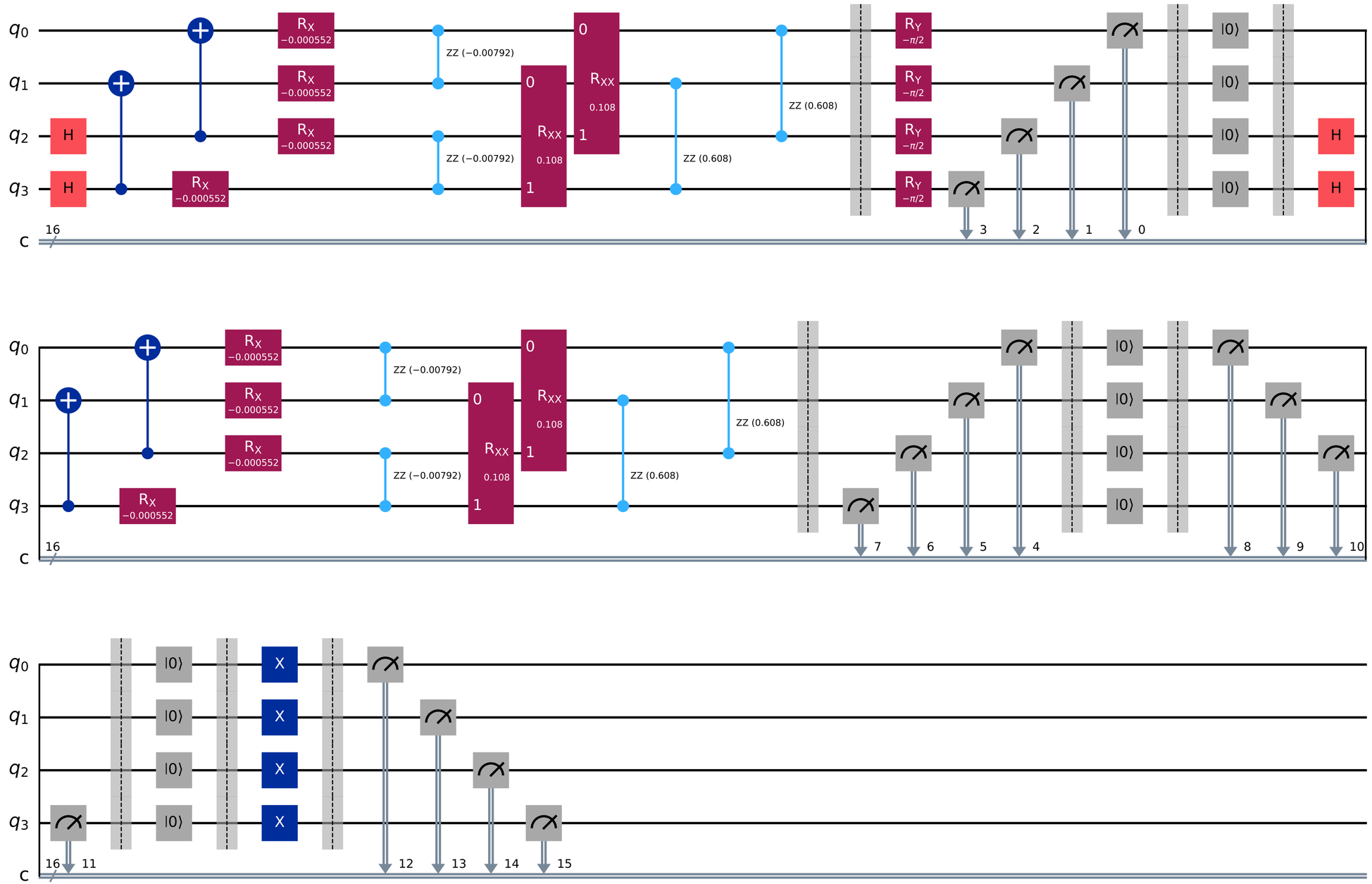

q_0
q_1
q_2
q_3
c
16
R_X −0.024
ZZ (−0.00661)
ZZ (0.6)
R_{XX} 0.0969
|0⟩
7
6
5
4
8
9
10
11

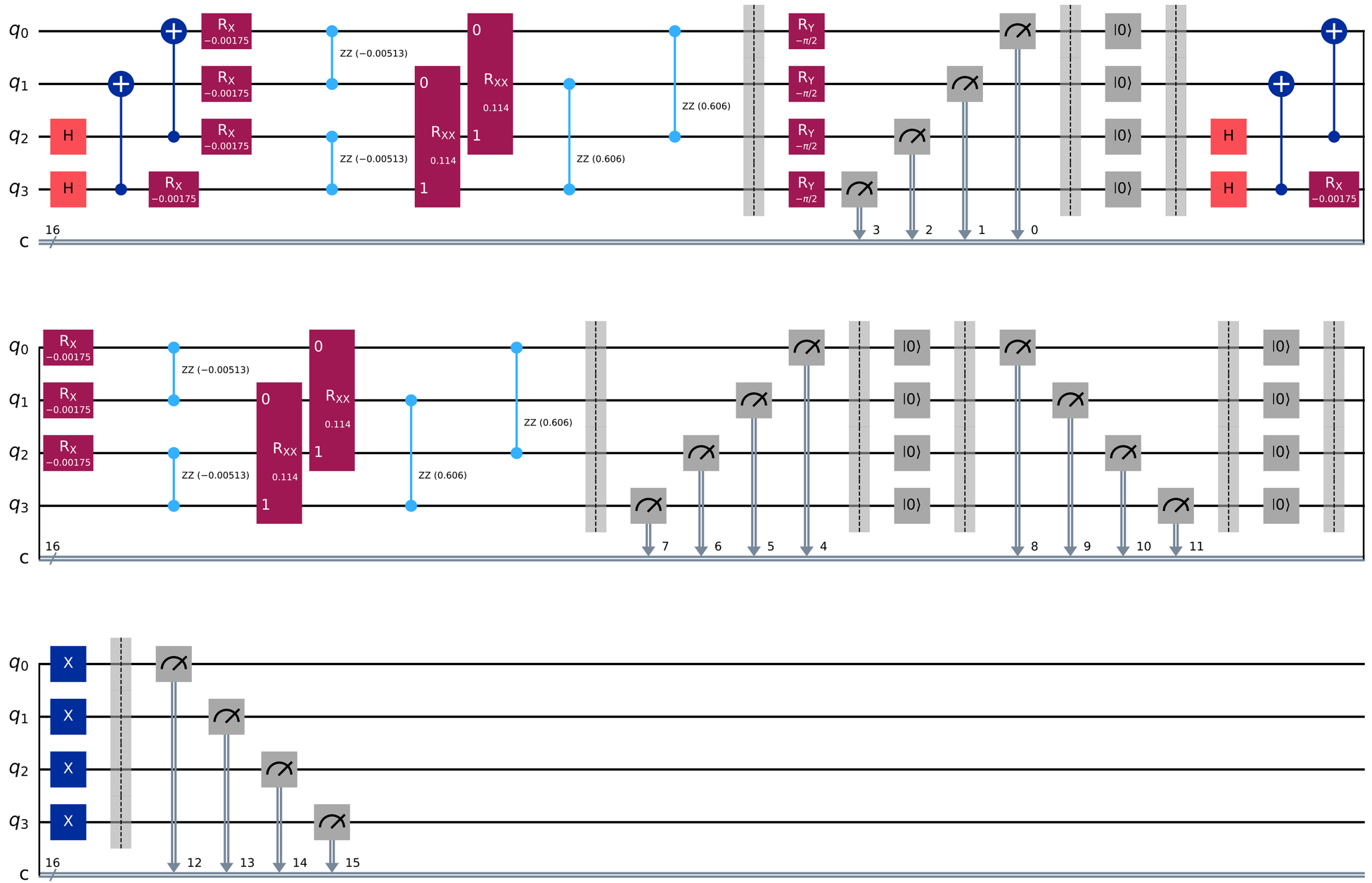

q_0
q_1
q_2
q_3
c
16
X
12
13
14
15

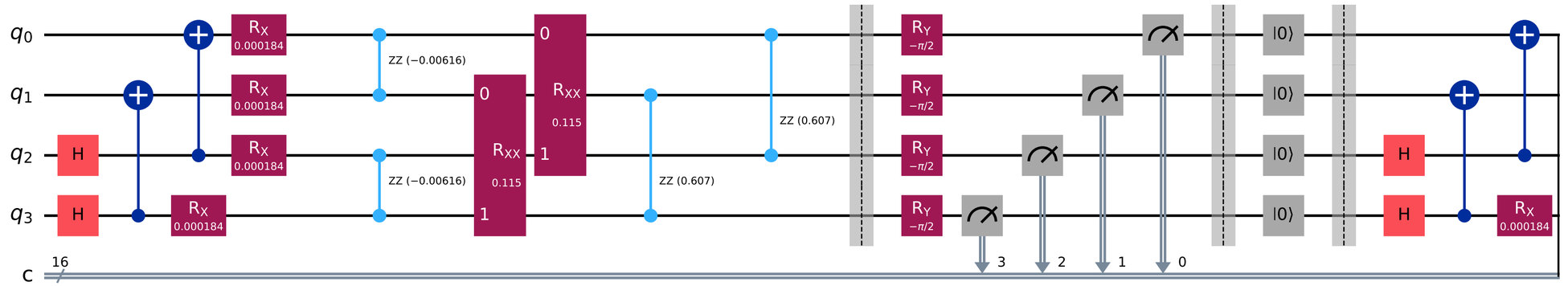

q_0
q_1
q_2
q_3
c
16
H
H
R_X −0.00386
ZZ (−0.0143)
ZZ (0.605)
R_{XX} 0.109
R_Y −π/2
|0⟩
X
3
2
1
0
7
6
5
4
8
9
10
11
12
13
14
15

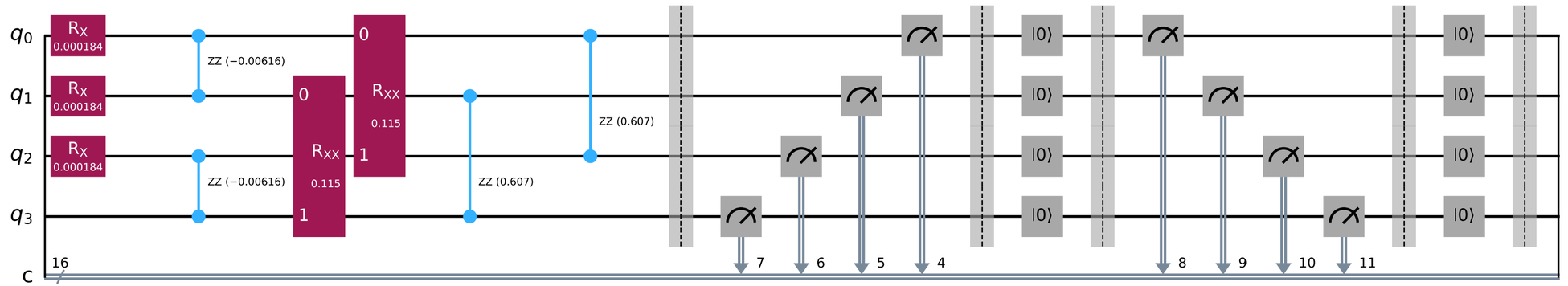
q_0
q_1
q_2
q_3
c
16
H
H
R_X −0.00485
R_X −0.00485
R_X −0.00485
R_X −0.00485
ZZ (−0.013)
ZZ (−0.013)
0
R_{XX}
0.109
1
0
R_{XX}
0.109
1
ZZ (0.608)
ZZ (0.608)
R_Y −π/2
R_Y −π/2
R_Y −π/2
R_Y −π/2
|0⟩
|0⟩
|0⟩
|0⟩
H
H
3
2
1
0
R_X −0.00485
R_X −0.00485
R_X −0.00485
R_X −0.00485
ZZ (−0.013)
ZZ (−0.013)
0
R_{XX}
0.109
1
0
R_{XX}
0.109
1
ZZ (0.608)
ZZ (0.608)
|0⟩
|0⟩
|0⟩
|0⟩
7
6
5
4
8
9
10
11
|0⟩
|0⟩
|0⟩
|0⟩
X
X
X
X
12
13
14
15

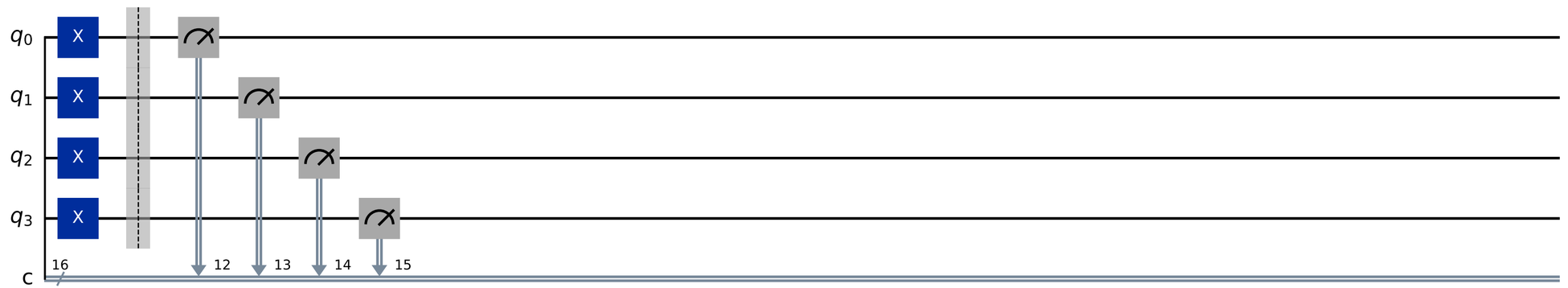

q_0
q_1
q_2
q_3
c
16
H
H
R_X −0.00151
ZZ (−0.00773)
ZZ (−0.00773)
R_{XX} 0.11
ZZ (0.607)
ZZ (0.607)
R_Y −π/2
|0⟩
X
3
2
1
0
7
6
5
4
8
9
10
11
12
13
14
15

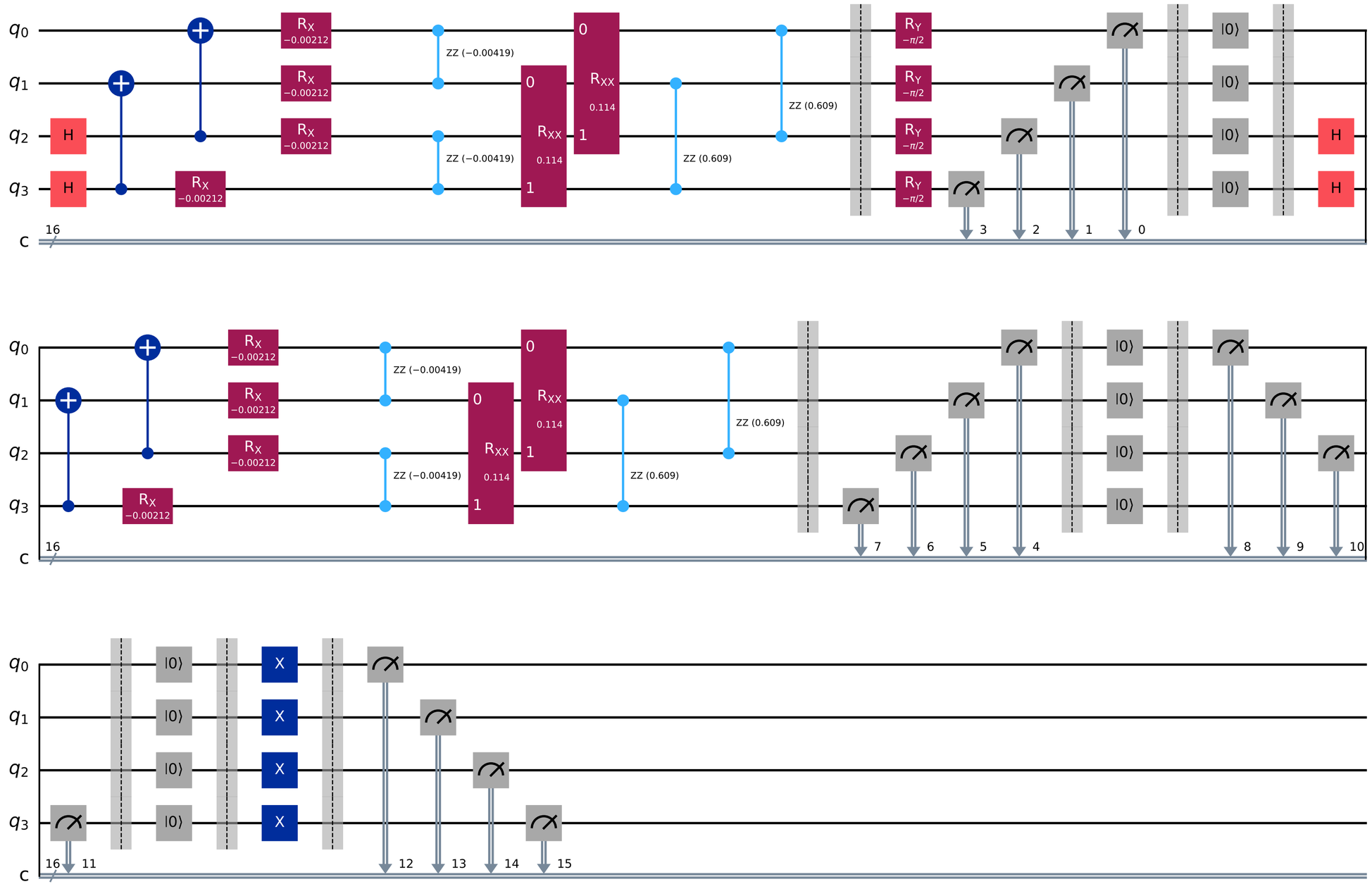

q_0
q_1
q_2
q_3
c
16
H
H
R_X −0.000552
R_X −0.000552
R_X −0.000552
R_X −0.000552
ZZ (−0.00792)
ZZ (−0.00792)
R_{XX} 0.108
ZZ (0.608)
ZZ (0.608)
R_Y −π/2
R_Y −π/2
R_Y −π/2
|0⟩
|0⟩
|0⟩
|0⟩
H
H
3
2
1
0
X
X
X
X
11
12
13
14
15
7
6
5
4
8
9
10

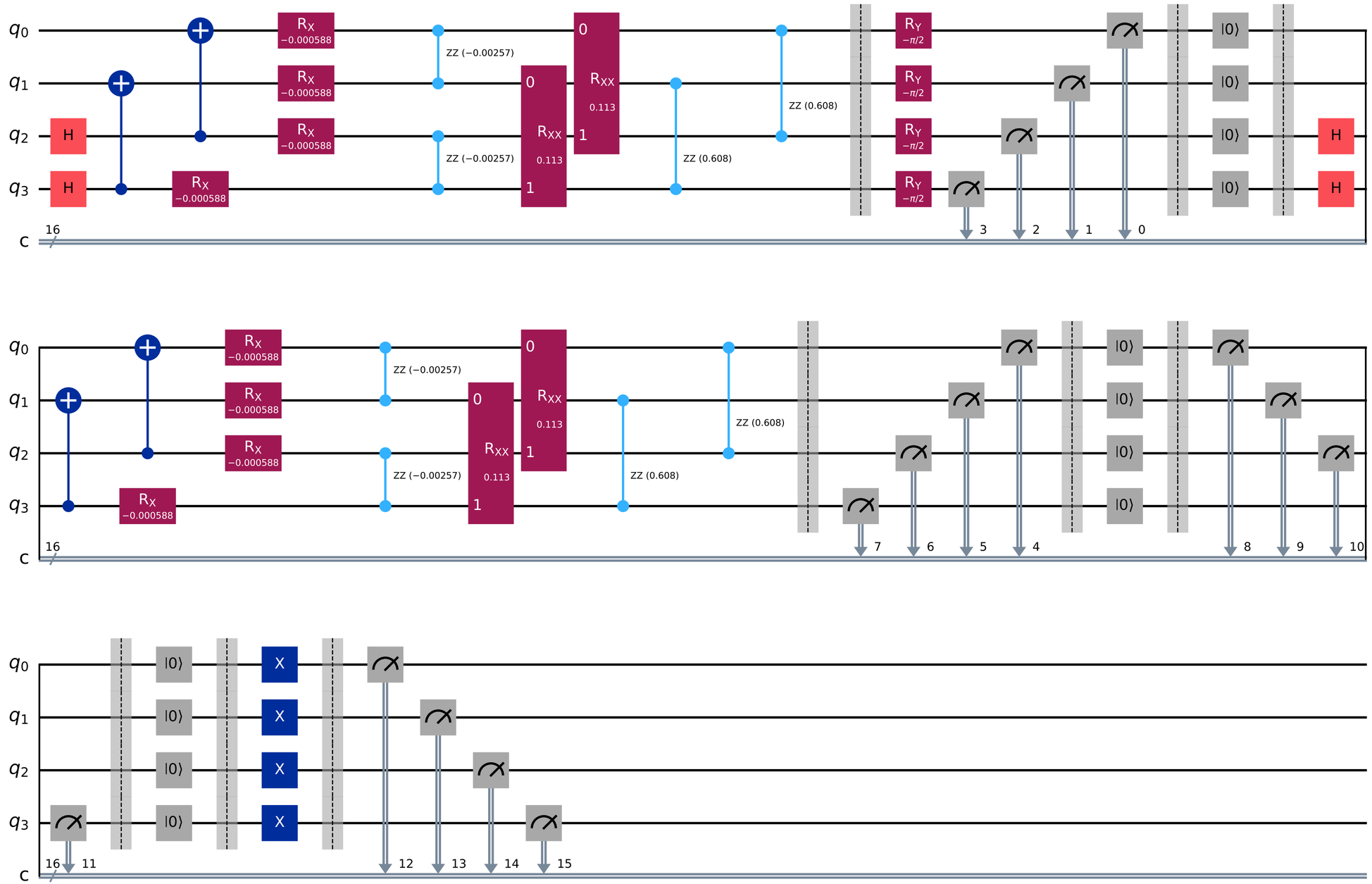

q_0
q_1
q_2
q_3
c
16
H
H
R_X −0.00175
ZZ (−0.00513)
ZZ (0.606)
R_{XX} 0.114
R_Y −π/2
|0⟩
X
0
1
3
2
1
0
7
6
5
4
8
9
10
11
12
13
14
15

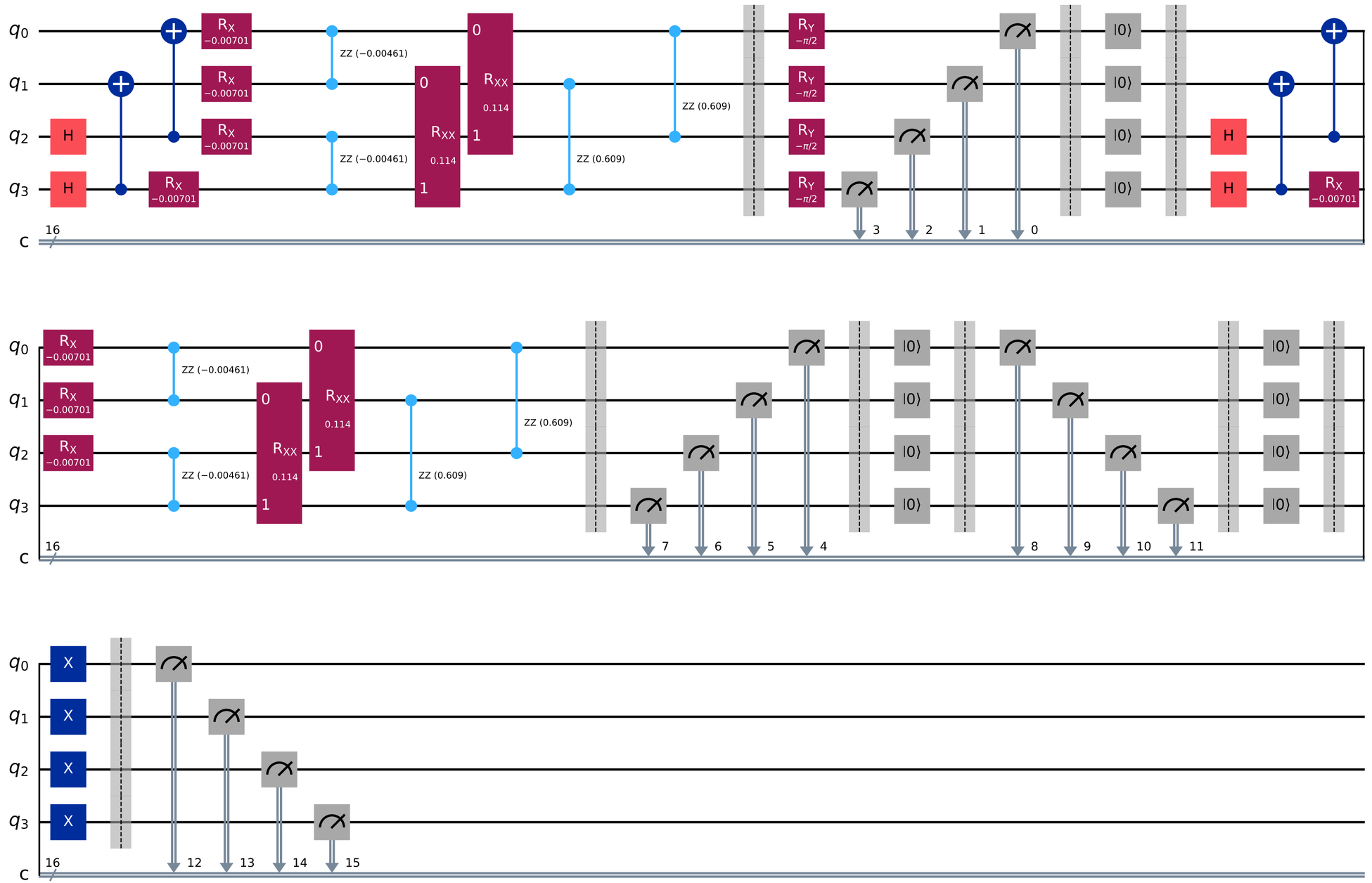

q_0
q_1
q_2
q_3
c
16
H
H
R_X 0.000184
R_X 0.000184
R_X 0.000184
R_X 0.000184
ZZ (−0.00616)
ZZ (−0.00616)
0
R_{XX} 0.115
1
0
R_{XX} 0.115
1
ZZ (0.607)
ZZ (0.607)
R_Y $-\pi/2$
R_Y $-\pi/2$
R_Y $-\pi/2$
R_Y $-\pi/2$
|0⟩
|0⟩
|0⟩
|0⟩
H
H
R_X 0.000184
3
2
1
0


q_0
q_1
q_2
q_3
c
16
R_X 0.000184
R_X 0.000184
R_X 0.000184
ZZ (−0.00616)
ZZ (−0.00616)
0
R_{XX} 0.115
1
0
R_{XX} 0.115
1
ZZ (0.607)
ZZ (0.607)
|0⟩
|0⟩
|0⟩
|0⟩
|0⟩
|0⟩
7
6
5
4
8
9
10
11

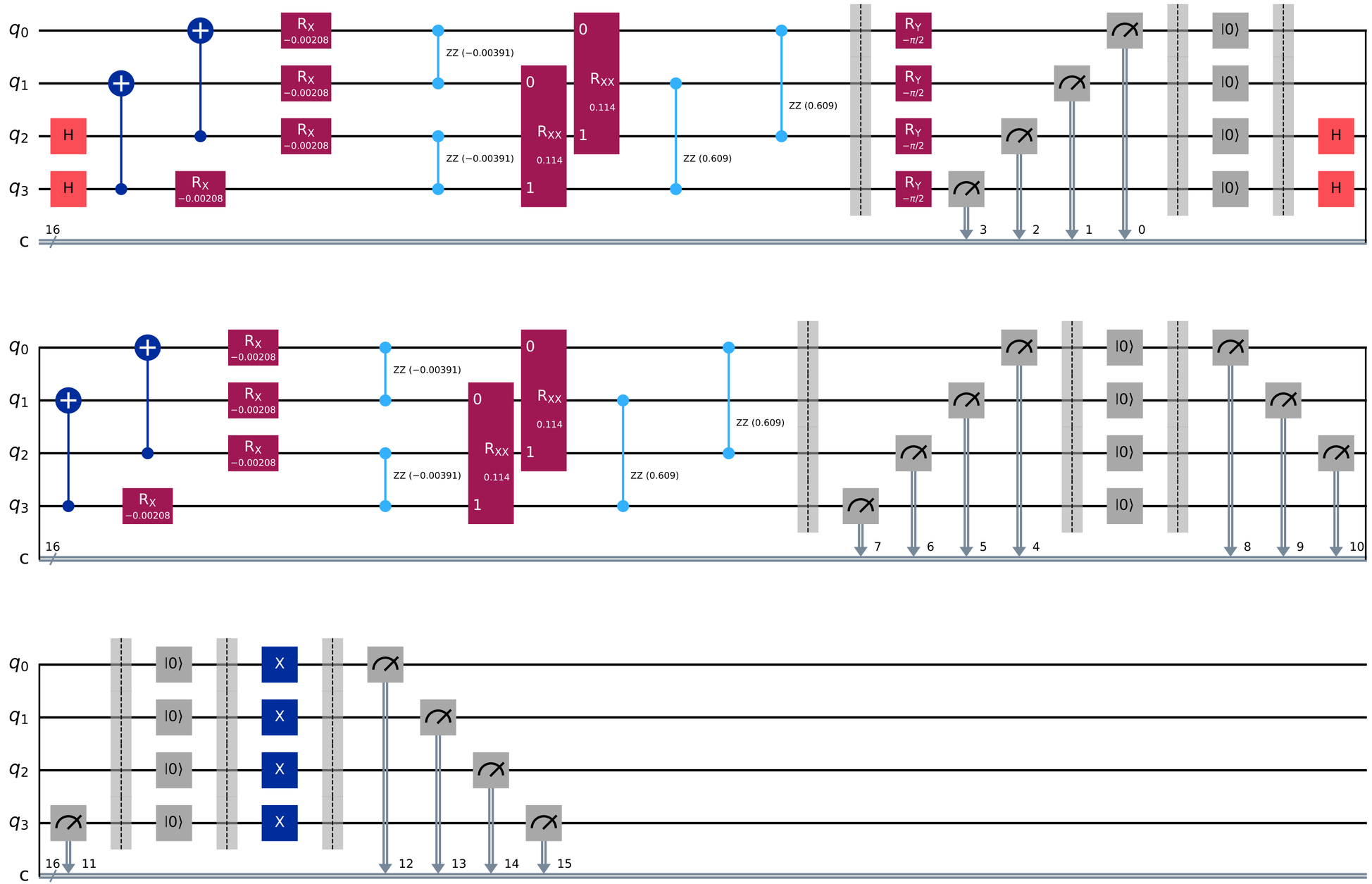

q_0
q_1
q_2
q_3
c
16
X
X
X
X
12
13
14
15

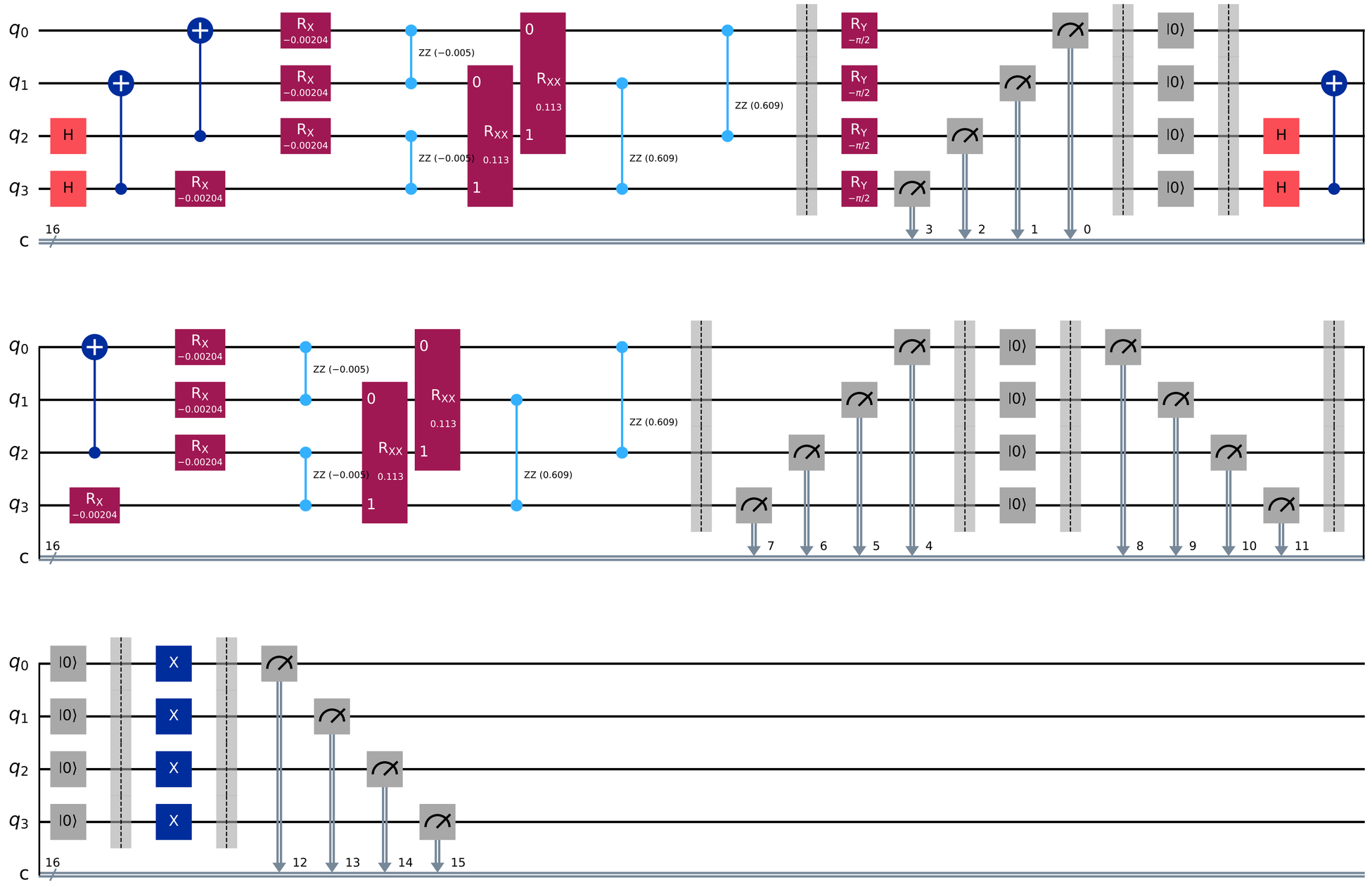

q_0
q_1
q_2
q_3
c
16
H
H
R_X −0.00212
R_X −0.00212
R_X −0.00212
R_X −0.00212
ZZ (−0.00419)
ZZ (−0.00419)
0
R_{XX}
0.114
1
0
R_{XX}
0.114
1
ZZ (0.609)
ZZ (0.609)
R_Y −π/2
R_Y −π/2
R_Y −π/2
|0⟩
|0⟩
|0⟩
|0⟩
H
H
3
2
1
0
R_X −0.00212
R_X −0.00212
R_X −0.00212
R_X −0.00212
ZZ (−0.00419)
ZZ (−0.00419)
0
R_{XX}
0.114
1
0
R_{XX}
0.114
1
ZZ (0.609)
ZZ (0.609)
|0⟩
|0⟩
|0⟩
|0⟩
7
6
5
4
8
9
10
|0⟩
|0⟩
|0⟩
|0⟩
X
X
X
X
16
11
12
13
14
15

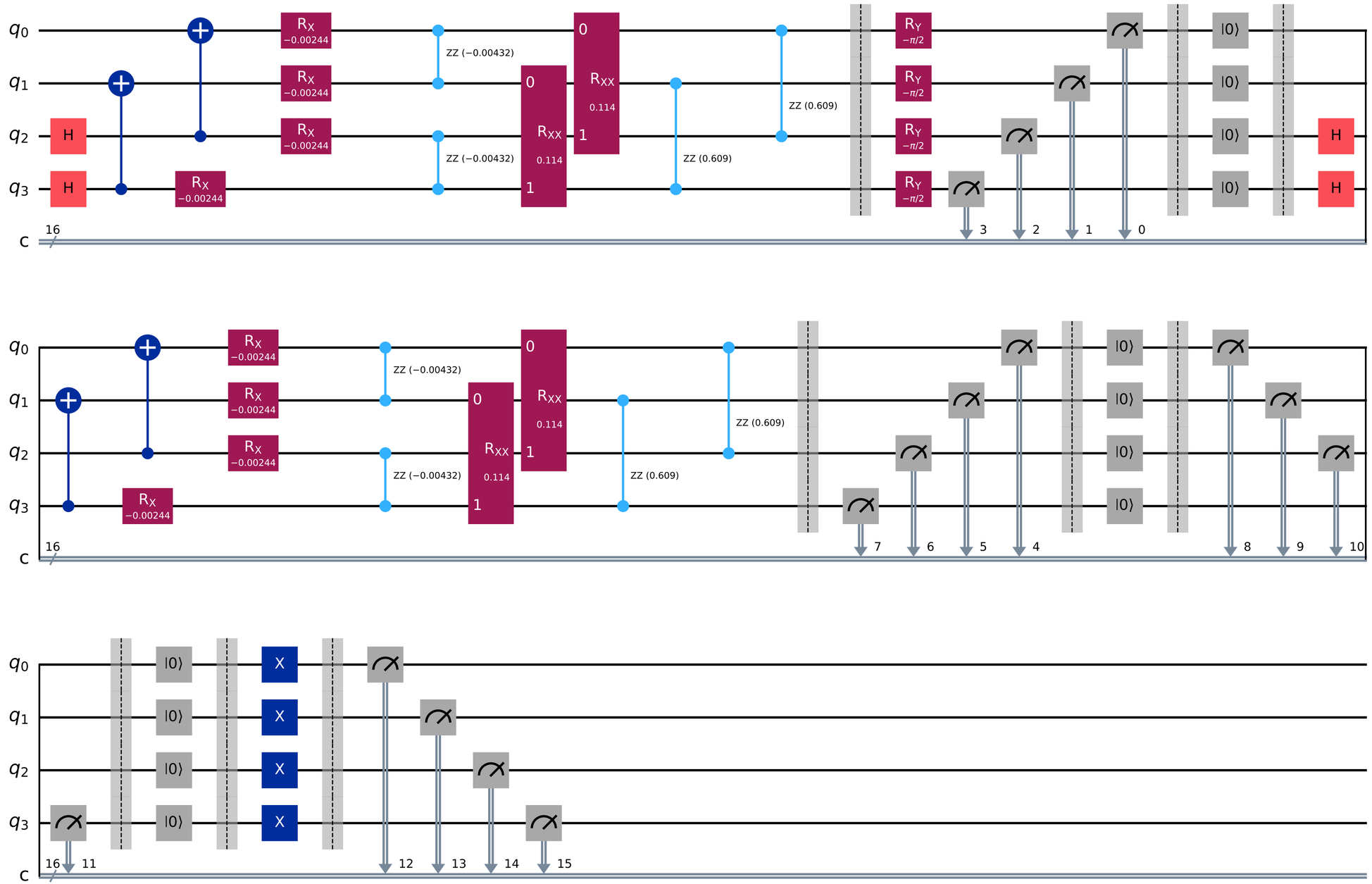

q_0
q_1
q_2
q_3
c
16
H
H
R_X −0.000588
R_X −0.000588
R_X −0.000588
R_X −0.000588
ZZ (−0.00257)
ZZ (−0.00257)
R_{XX} 0.113
ZZ (0.608)
ZZ (0.608)
R_Y −π/2
R_Y −π/2
R_Y −π/2
R_Y −π/2
|0⟩
|0⟩
|0⟩
|0⟩
H
H
3
2
1
0
X
X
X
X
7
6
5
4
11
12
13
14
15
8
9
10

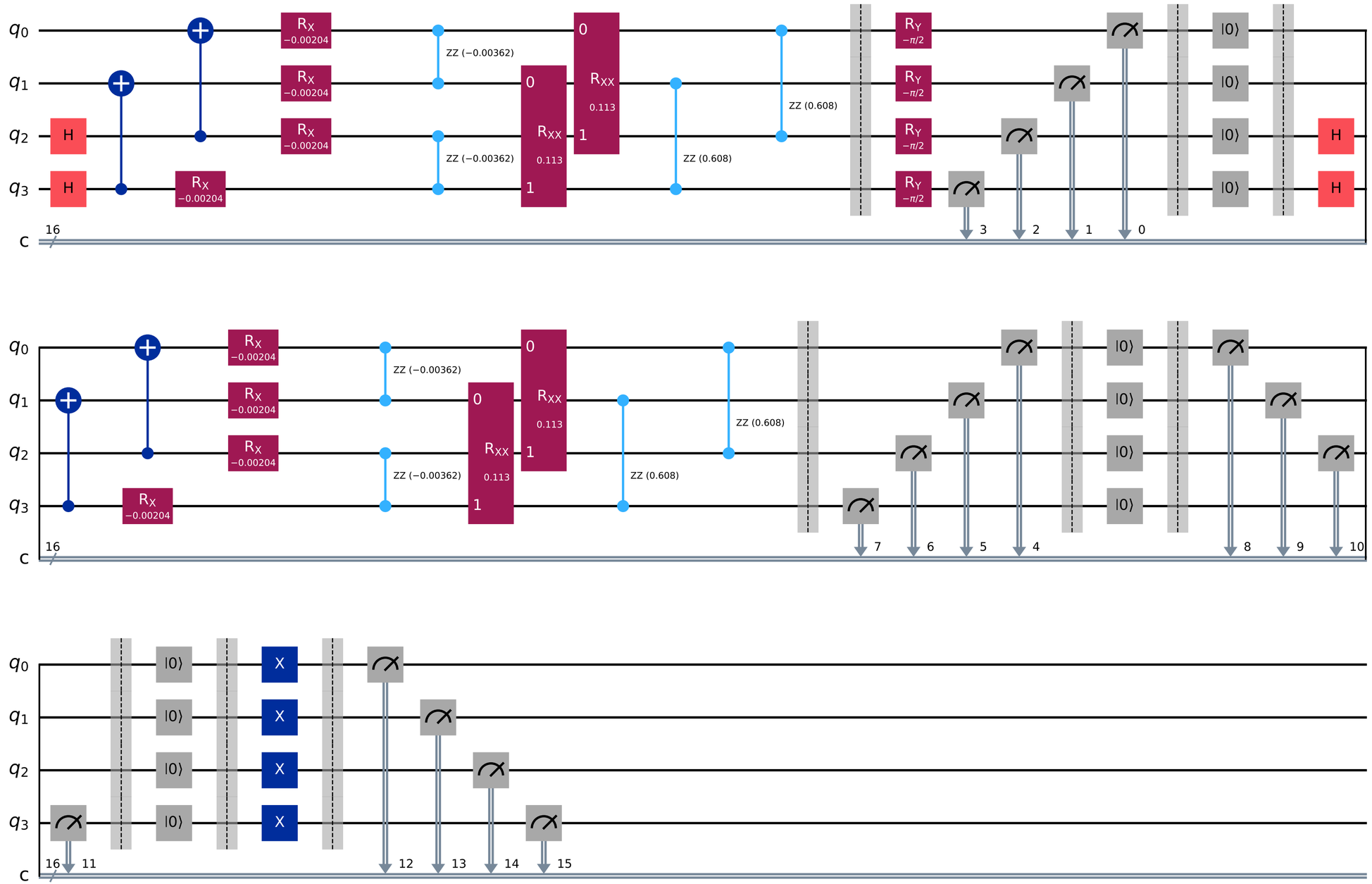

q_0
q_1
q_2
q_3
c
16
H
H
R_X −0.00701
R_X −0.00701
R_X −0.00701
R_X −0.00701
ZZ (−0.00461)
ZZ (−0.00461)
R_{XX} 0.114
ZZ (0.609)
ZZ (0.609)
R_Y −π/2
R_Y −π/2
R_Y −π/2
R_Y −π/2
|0⟩
|0⟩
|0⟩
|0⟩
H
H
R_X −0.00701
3
2
1
0
R_X −0.00701
R_X −0.00701
R_X −0.00701
ZZ (−0.00461)
ZZ (−0.00461)
R_{XX} 0.114
ZZ (0.609)
ZZ (0.609)
|0⟩
|0⟩
|0⟩
|0⟩
|0⟩
7
6
5
4
8
9
10
11
X
X
X
X
12
13
14
15

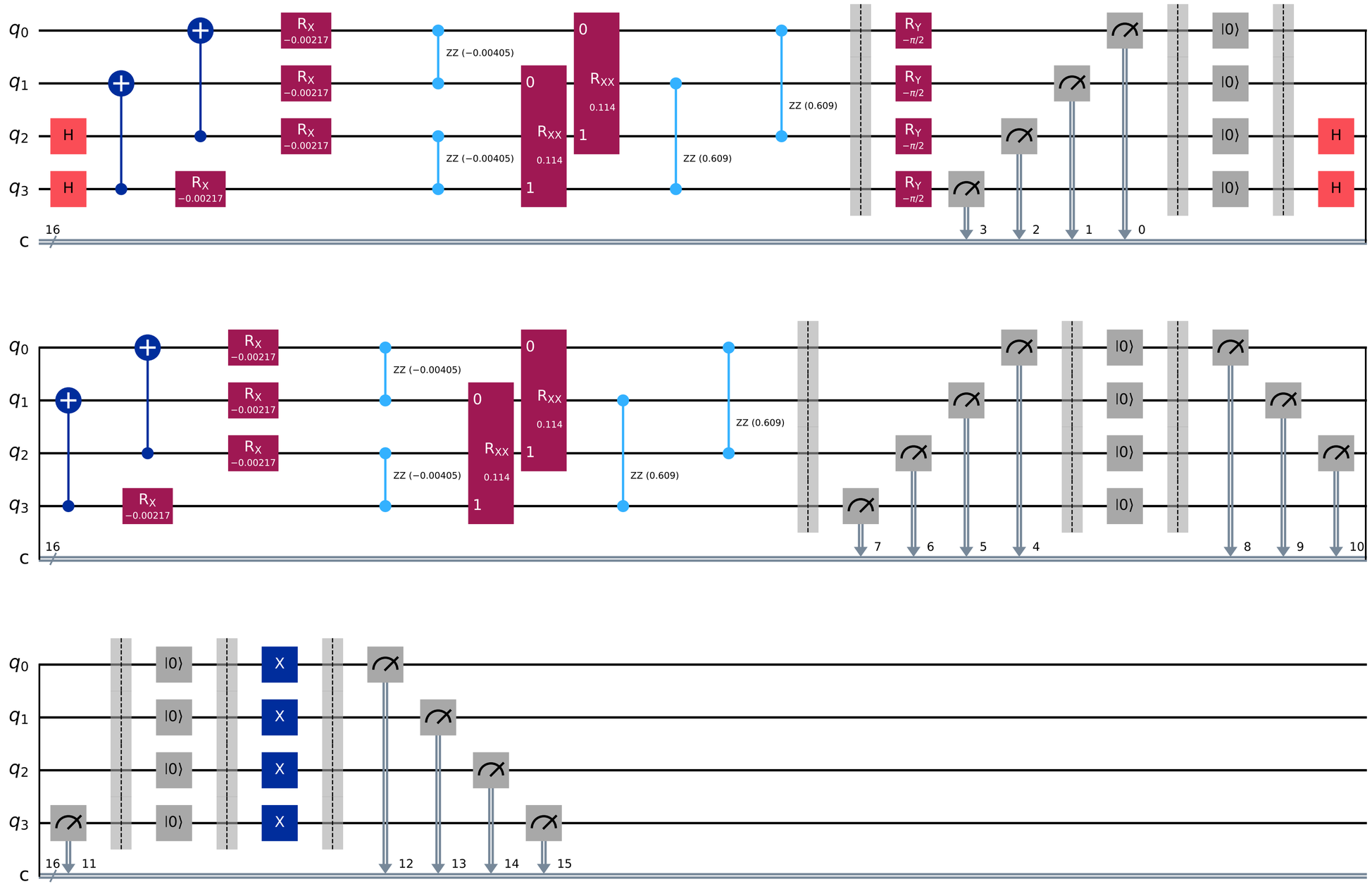
q_0
q_1
q_2
q_3
c
16
H
H
R_X −0.002
ZZ (−0.00202)
ZZ (0.608)
R_{XX} 0.113
R_Y −π/2
|0⟩
X
3
2
1
0
7
6
5
4
8
9
10
11
12
13
14
15

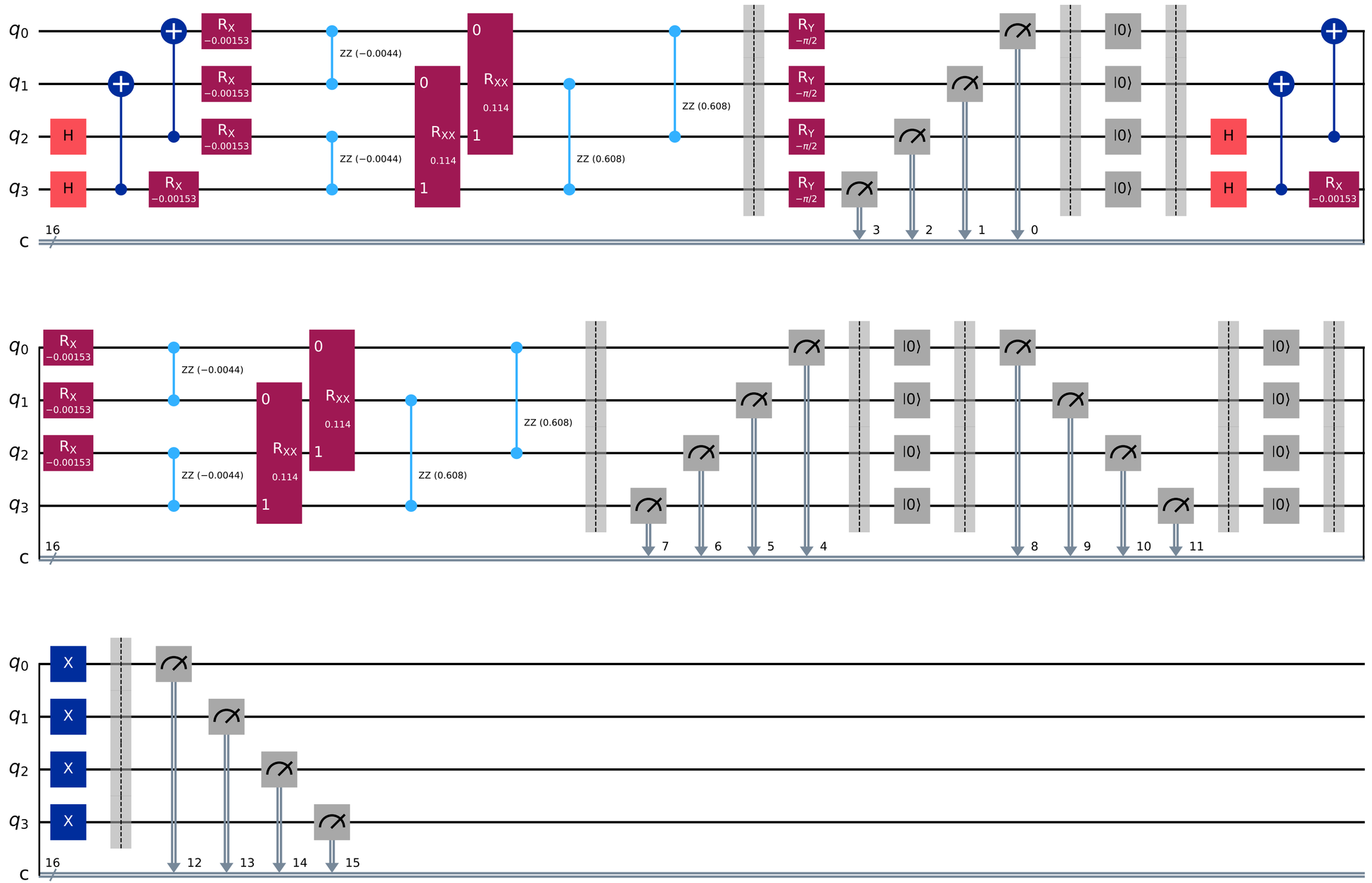

q_0
q_1
q_2
q_3
c
16
H
H
R_X −0.00208
R_X −0.00208
R_X −0.00208
R_X −0.00208
ZZ (−0.00391)
ZZ (−0.00391)
0
R_{XX}
0.114
1
0
R_{XX}
0.114
1
ZZ (0.609)
ZZ (0.609)
R_Y −π/2
R_Y −π/2
R_Y −π/2
R_Y −π/2
|0⟩
|0⟩
|0⟩
|0⟩
H
H
3
2
1
0
R_X −0.00208
R_X −0.00208
R_X −0.00208
R_X −0.00208
ZZ (−0.00391)
ZZ (−0.00391)
0
R_{XX}
0.114
1
0
R_{XX}
0.114
1
ZZ (0.609)
ZZ (0.609)
|0⟩
|0⟩
|0⟩
|0⟩
7
6
5
4
8
9
10
X
X
X
X
|0⟩
|0⟩
|0⟩
|0⟩
11
12
13
14
15

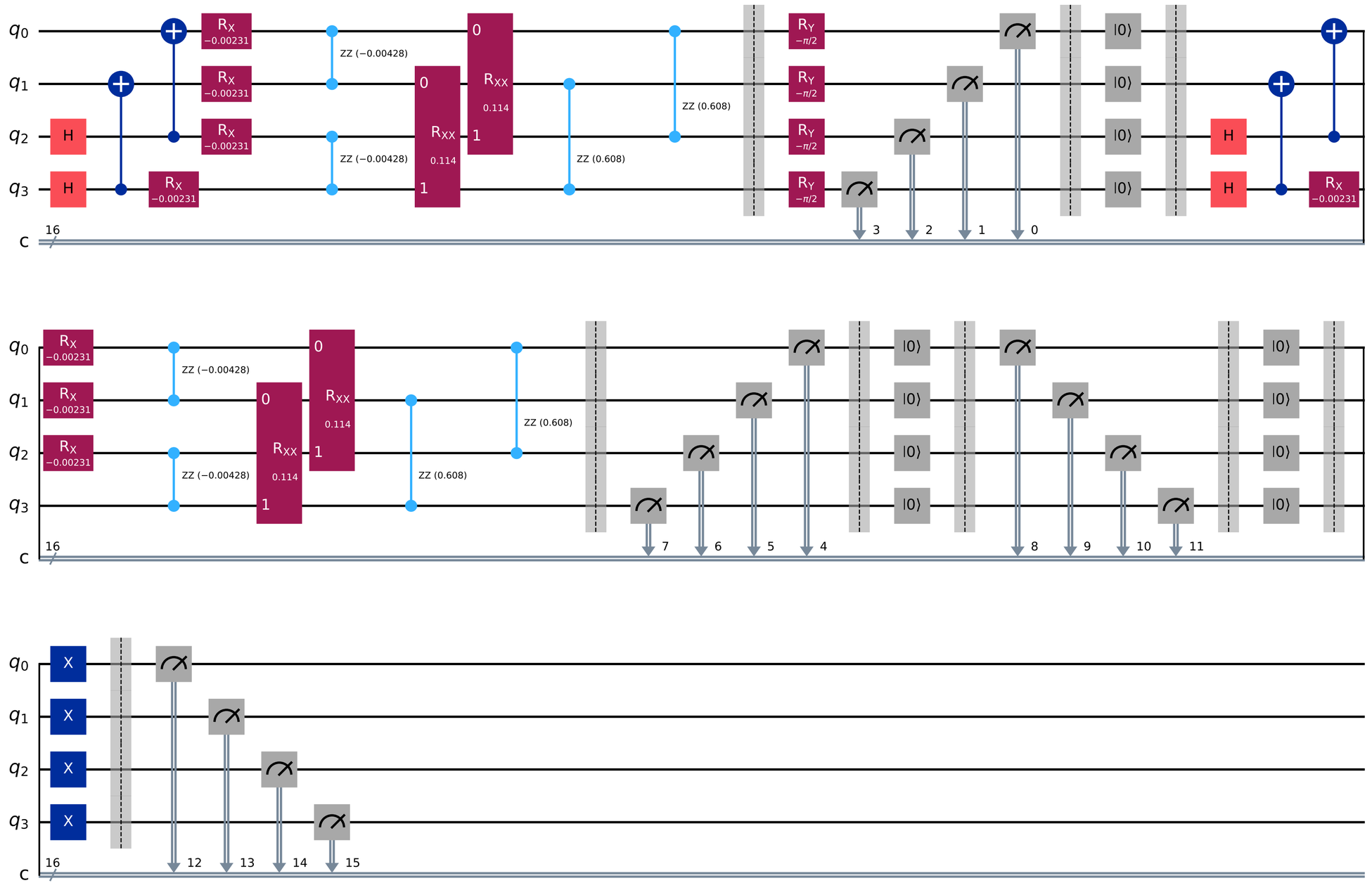

q_0
q_1
q_2
q_3
c
16
H
H
R_X −0.00204
R_X −0.00204
R_X −0.00204
R_X −0.00204
ZZ (−0.005)
ZZ (−0.005)
0
R_{XX}
1
0.113
0
R_{XX}
1
0.113
ZZ (0.609)
ZZ (0.609)
R_Y −π/2
R_Y −π/2
R_Y −π/2
R_Y −π/2
|0⟩
|0⟩
|0⟩
|0⟩
H
H
3
2
1
0
R_X −0.00204
R_X −0.00204
R_X −0.00204
R_X −0.00204
ZZ (−0.005)
ZZ (−0.005)
0
R_{XX}
1
0.113
0
R_{XX}
1
0.113
ZZ (0.609)
ZZ (0.609)
|0⟩
|0⟩
|0⟩
|0⟩
7
6
5
4
8
9
10
11
|0⟩
|0⟩
|0⟩
|0⟩
X
X
X
X
12
13
14
15

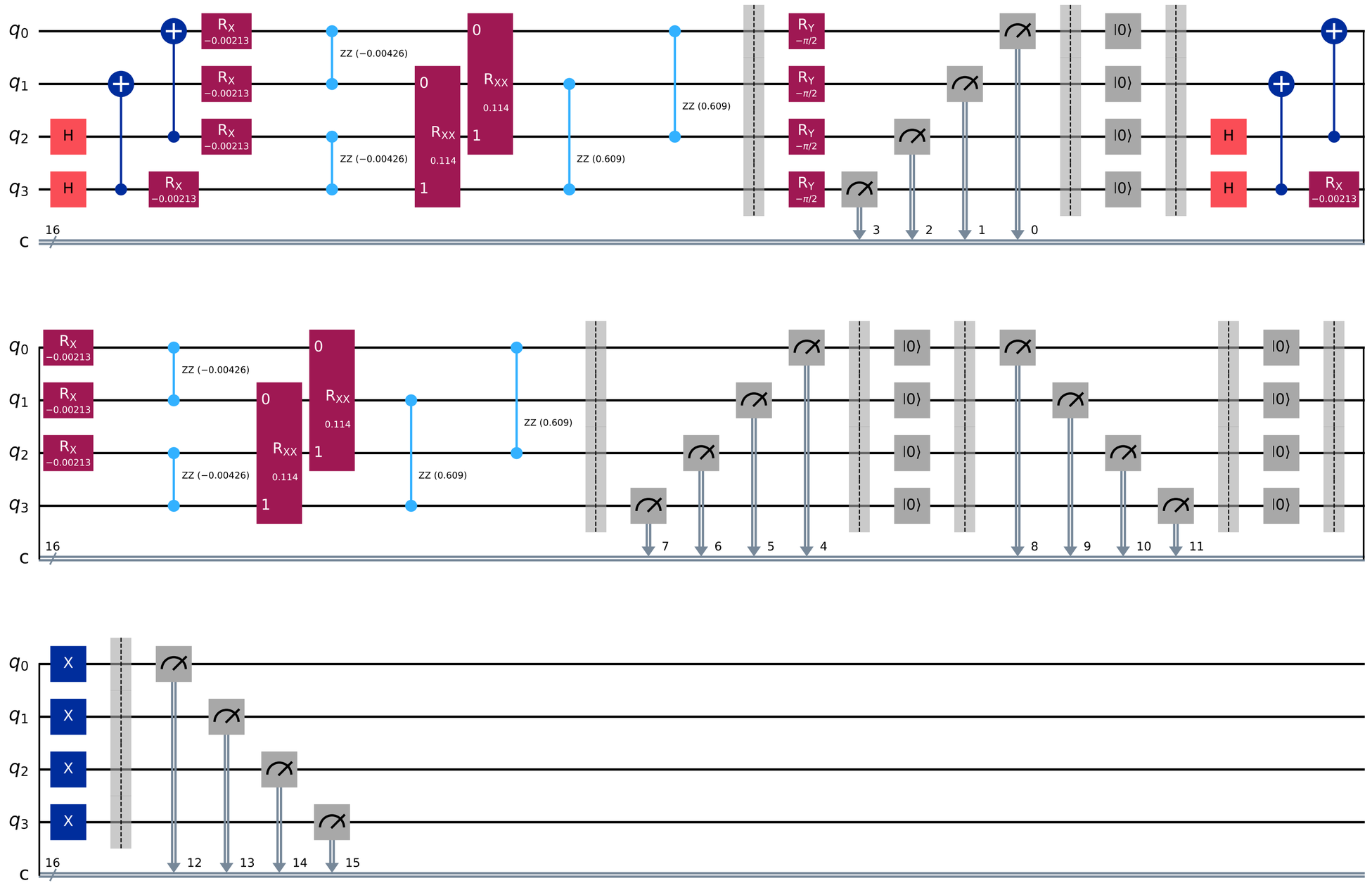

q_0
q_1
q_2
q_3
c
16
H
H
R_X −0.00244
R_X −0.00244
R_X −0.00244
R_X −0.00244
ZZ (−0.00432)
ZZ (−0.00432)
0
R_{XX}
0.114
1
0
R_{XX}
0.114
1
ZZ (0.609)
ZZ (0.609)
R_Y −π/2
R_Y −π/2
R_Y −π/2
|0⟩
|0⟩
|0⟩
|0⟩
H
H
3
2
1
0
R_X −0.00244
R_X −0.00244
R_X −0.00244
R_X −0.00244
ZZ (−0.00432)
ZZ (−0.00432)
0
R_{XX}
0.114
1
0
R_{XX}
0.114
1
ZZ (0.609)
ZZ (0.609)
|0⟩
|0⟩
|0⟩
|0⟩
7
6
5
4
8
9
10
|0⟩
|0⟩
|0⟩
|0⟩
X
X
X
X
16
11
12
13
14
15

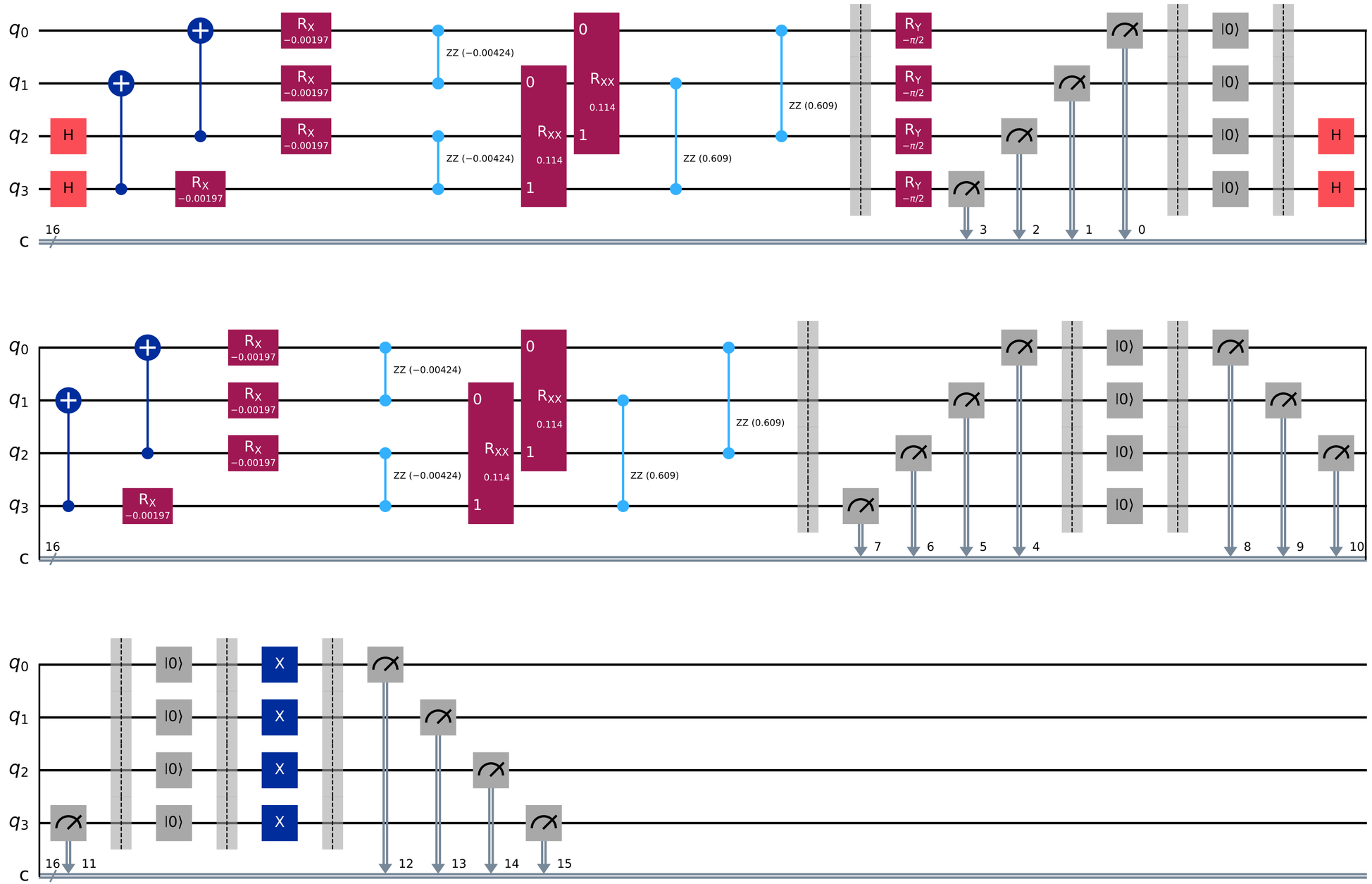

q_0
q_1
q_2
q_3
c
16
H
H
R_X −0.00204
R_X −0.00204
R_X −0.00204
R_X −0.00204
ZZ (−0.00362)
ZZ (−0.00362)
0
R_{XX}
0.113
1
0
R_{XX}
0.113
1
ZZ (0.608)
ZZ (0.608)
R_Y −π/2
R_Y −π/2
R_Y −π/2
|0⟩
|0⟩
|0⟩
|0⟩
H
H
3
2
1
0
R_X −0.00204
R_X −0.00204
R_X −0.00204
R_X −0.00204
ZZ (−0.00362)
ZZ (−0.00362)
0
R_{XX}
0.113
1
0
R_{XX}
0.113
1
ZZ (0.608)
ZZ (0.608)
|0⟩
|0⟩
|0⟩
|0⟩
7
6
5
4
8
9
10
X
X
X
X
|0⟩
|0⟩
|0⟩
|0⟩
11
12
13
14
15

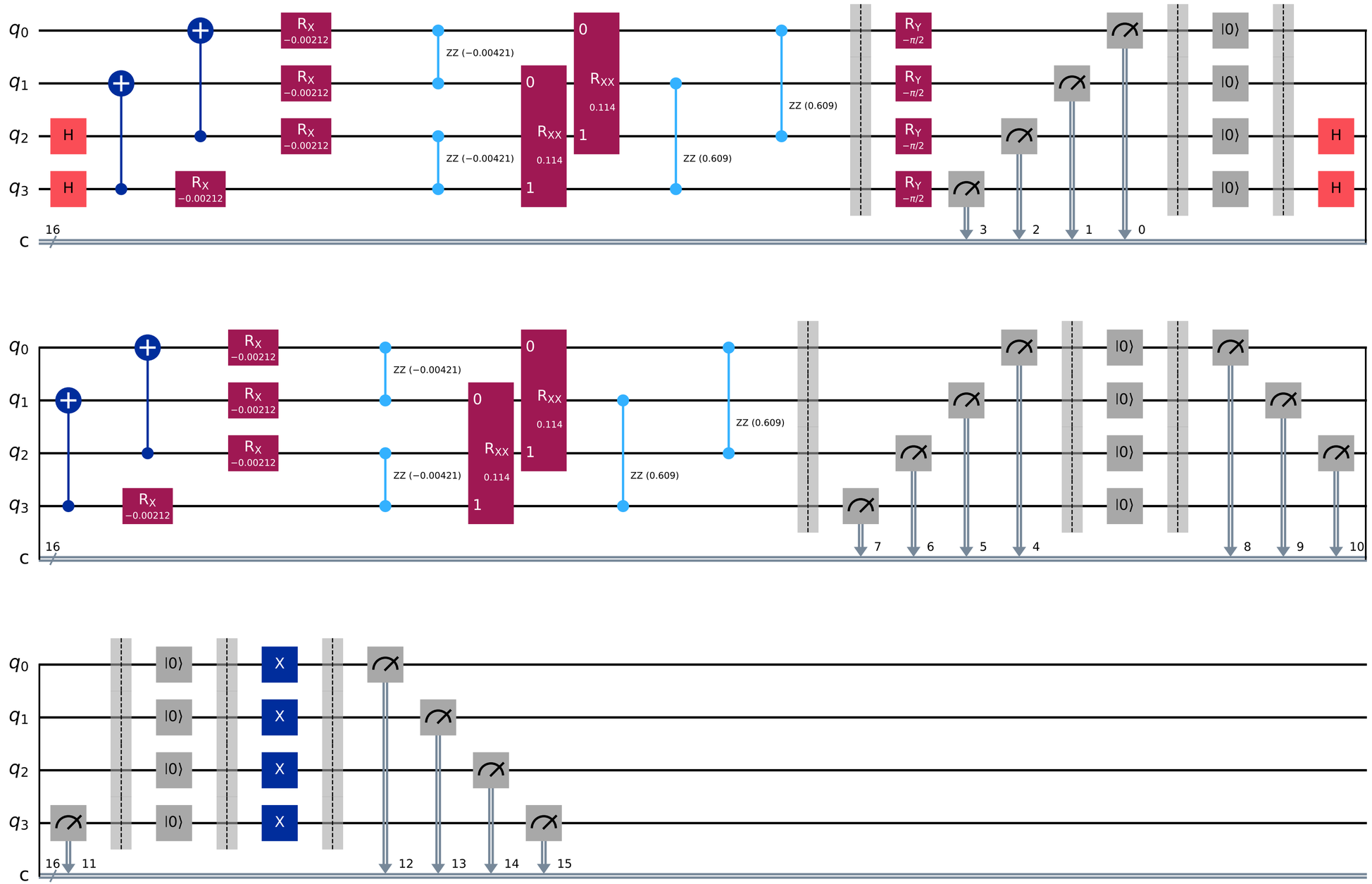

q_0
q_1
q_2
q_3
c
16
H
H
R_X −0.00217
R_X −0.00217
R_X −0.00217
R_X −0.00217
ZZ (−0.00405)
ZZ (−0.00405)
0
R_{XX}
0.114
1
0
R_{XX}
0.114
1
ZZ (0.609)
ZZ (0.609)
R_Y −π/2
R_Y −π/2
R_Y −π/2
|0⟩
|0⟩
|0⟩
|0⟩
H
H
3
2
1
0
R_X −0.00217
R_X −0.00217
R_X −0.00217
R_X −0.00217
ZZ (−0.00405)
ZZ (−0.00405)
0
R_{XX}
0.114
1
0
R_{XX}
0.114
1
ZZ (0.609)
ZZ (0.609)
|0⟩
|0⟩
|0⟩
|0⟩
7
6
5
4
8
9
10
X
X
X
X
11
12
13
14
15

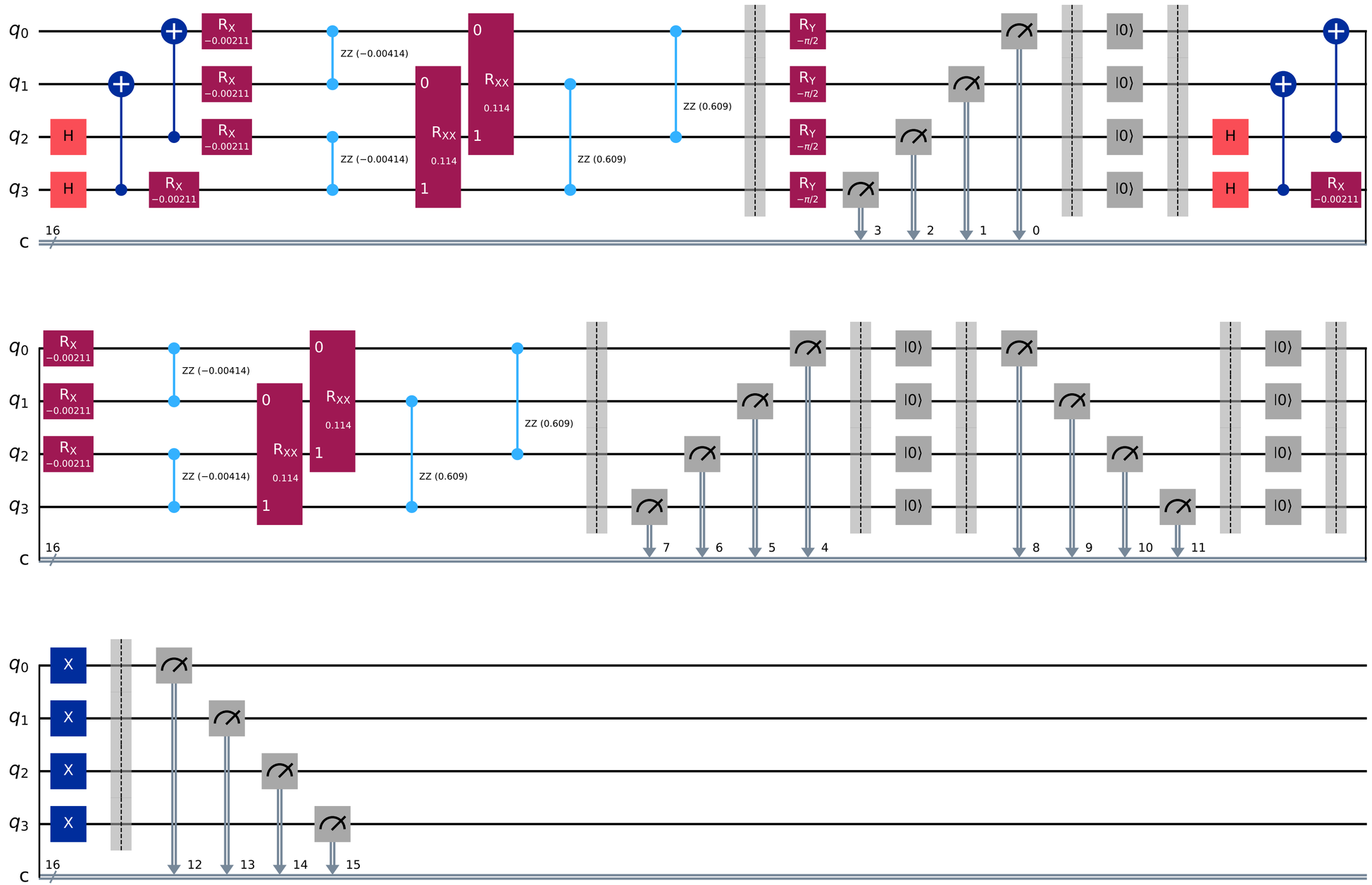

q_0
q_1
q_2
q_3
c
16
H
H
R_X −0.00153
ZZ (−0.0044)
ZZ (0.608)
R_{XX} 0.114
R_Y −π/2
|0⟩
X
3
2
1
0
7
6
5
4
8
9
10
11
12
13
14
15

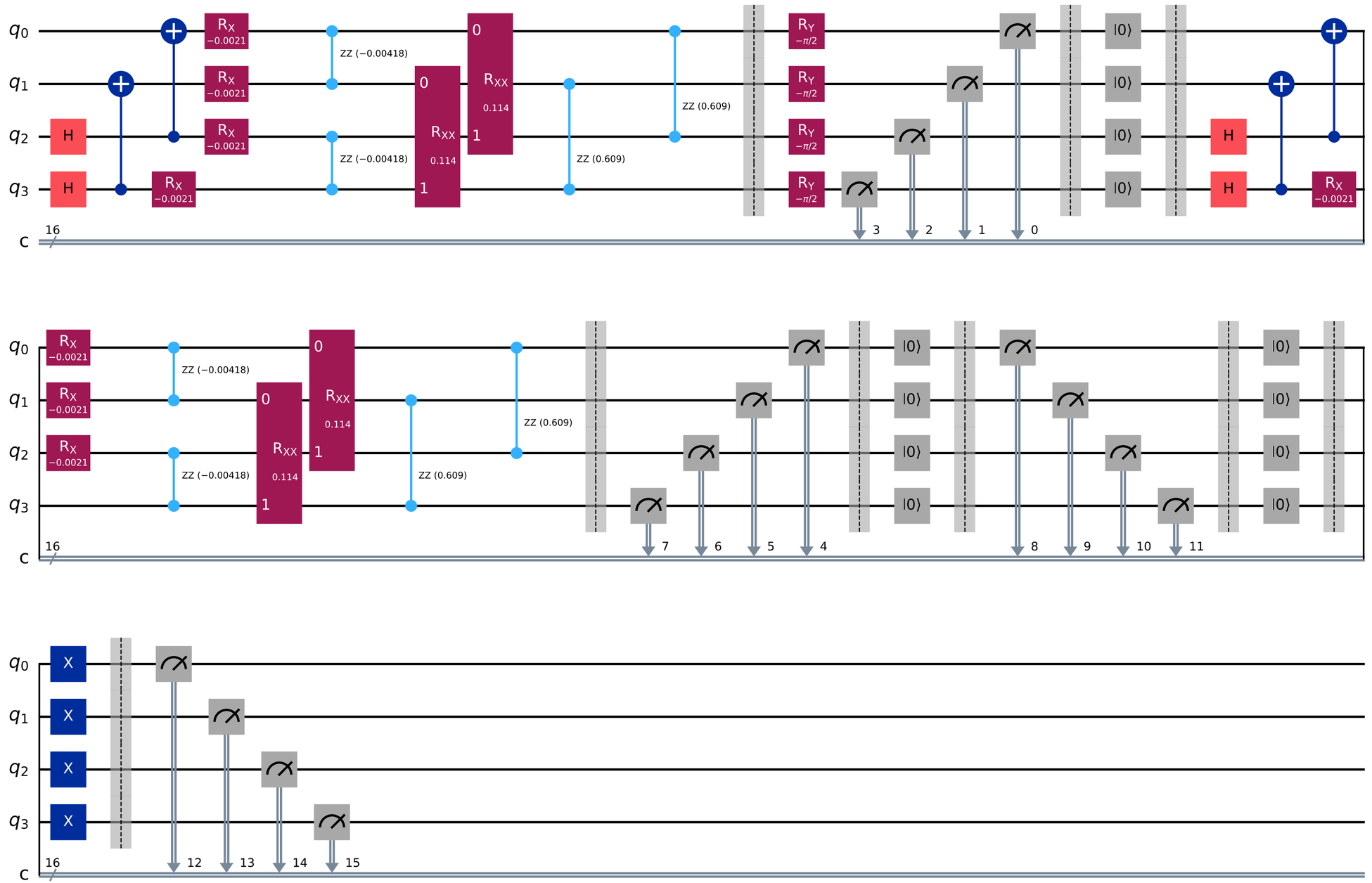

q_0
q_1
q_2
q_3
c
16
H
H
R_X −0.00231
R_X −0.00231
R_X −0.00231
R_X −0.00231
ZZ (−0.00428)
ZZ (−0.00428)
R_{XX} 0.114
ZZ (0.608)
ZZ (0.608)
R_Y −π/2
R_Y −π/2
R_Y −π/2
R_Y −π/2
|0⟩
|0⟩
|0⟩
|0⟩
H
H
R_X −0.00231
3
2
1
0
R_X −0.00231
R_X −0.00231
R_X −0.00231
ZZ (−0.00428)
ZZ (−0.00428)
R_{XX} 0.114
ZZ (0.608)
ZZ (0.608)
|0⟩
|0⟩
|0⟩
|0⟩
7
6
5
4
8
9
10
11
X
X
X
X
12
13
14
15

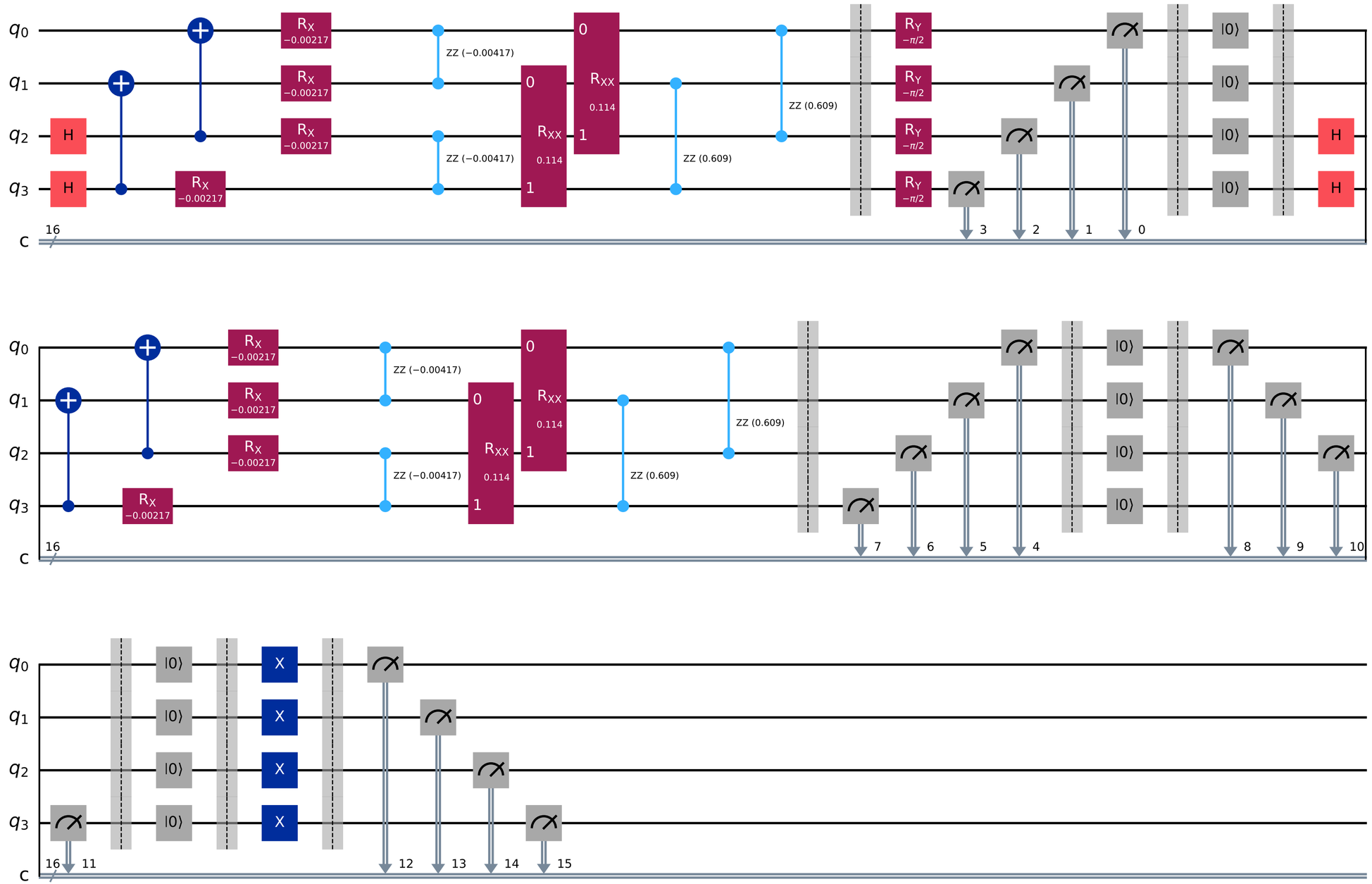

q0
q1
q2
q3
c
16
H
H
RX −0.00213
ZZ (−0.00426)
RXX 0.114
ZZ (0.609)
RY −π/2
|0⟩
3
2
1
0
7
6
5
4
8
9
10
11
12
13
14
15
X


q_0
q_1
q_2
q_3
c
16
H
H
R_X −0.00197
R_X −0.00197
R_X −0.00197
R_X −0.00197
ZZ (−0.00424)
ZZ (−0.00424)
0
R_{XX}
0.114
1
0
R_{XX}
0.114
1
ZZ (0.609)
ZZ (0.609)
R_Y −π/2
R_Y −π/2
R_Y −π/2
|0⟩
|0⟩
|0⟩
|0⟩
H
H
3
2
1
0
X
X
X
X
7
6
5
4
8
9
10
11
12
13
14
15


q_0
q_1
q_2
q_3
c
16
H
H
R_X −0.00212
R_X −0.00212
R_X −0.00212
R_X −0.00212
ZZ (−0.00421)
ZZ (−0.00421)
0
R_{XX}
0.114
1
0
R_{XX}
0.114
1
ZZ (0.609)
ZZ (0.609)
R_Y −π/2
R_Y −π/2
R_Y −π/2
|0⟩
|0⟩
|0⟩
|0⟩
H
H
3
2
1
0
R_X −0.00212
R_X −0.00212
R_X −0.00212
R_X −0.00212
ZZ (−0.00421)
ZZ (−0.00421)
0
R_{XX}
0.114
1
0
R_{XX}
0.114
1
ZZ (0.609)
ZZ (0.609)
|0⟩
|0⟩
|0⟩
|0⟩
7
6
5
4
8
9
10
|0⟩
|0⟩
|0⟩
|0⟩
X
X
X
X
16
11
12
13
14
15


q0
q1
q2
q3
c
16
H
H
RX −0.00211
RX −0.00211
RX −0.00211
RX −0.00211
ZZ (−0.00414)
ZZ (−0.00414)
0
RXX
0.114
1
0
RXX
0.114
1
ZZ (0.609)
ZZ (0.609)
RY −π/2
RY −π/2
RY −π/2
RY −π/2
|0⟩
|0⟩
|0⟩
|0⟩
H
H
RX −0.00211
3
2
1
0
RX −0.00211
RX −0.00211
RX −0.00211
ZZ (−0.00414)
ZZ (−0.00414)
0
RXX
0.114
1
0
RXX
0.114
1
ZZ (0.609)
ZZ (0.609)
|0⟩
|0⟩
|0⟩
|0⟩
|0⟩
|0⟩
|0⟩
|0⟩
7
6
5
4
8
9
10
11
X
X
X
X
12
13
14
15


q_0
q_1
q_2
q_3
c
16
H
H
R_X −0.0021
ZZ (−0.00418)
R_{XX} 0.114
ZZ (0.609)
R_Y −π/2
|0⟩
X
3
2
1
0
7
6
5
4
8
9
10
11
12
13
14
15


q_0
q_1
q_2
q_3
c
16
H
H
R_X −0.00217
R_X −0.00217
R_X −0.00217
R_X −0.00217
ZZ (−0.00417)
ZZ (−0.00417)
0
R_{XX}
0.114
1
0
R_{XX}
0.114
1
ZZ (0.609)
ZZ (0.609)
R_Y −π/2
R_Y −π/2
R_Y −π/2
|0⟩
|0⟩
|0⟩
|0⟩
H
H
3
2
1
0
R_X −0.00217
R_X −0.00217
R_X −0.00217
R_X −0.00217
ZZ (−0.00417)
ZZ (−0.00417)
0
R_{XX}
0.114
1
0
R_{XX}
0.114
1
ZZ (0.609)
ZZ (0.609)
|0⟩
|0⟩
|0⟩
|0⟩
7
6
5
4
8
9
10
|0⟩
|0⟩
|0⟩
|0⟩
X
X
X
X
11
12
13
14
15